Simplified Medical Reports

Clinical review pack: original reports and AI-written plain-language versions

Why we are asking

We are testing whether AI language models can rewrite medical reports into plain language that patients and their families can understand. The models were asked to keep every clinical fact, explain each medical term in simple words, and write at roughly a 6th-grade reading level in plain Indian English. Your judgement on accuracy and safety is the part that automatic measures cannot provide.

What is in this pack

15 original reports (discharge summaries, prostate pathology reports and brain MRI reports), each followed by its simplified versions: 29 versions in total. Each version is labelled with a letter (A, B, C…). The letters are assigned in random order and do not reveal which AI method produced the text, so that each version is judged on its own.

For each version, please check

- **Accuracy**: is anything wrong, misinterpreted, or placed at the wrong time?
- **Invented information**: does it state anything that is not in the original (for example test values, warning signs, instructions or relationships)?
- **Missing information**: is anything clinically important left out?
- **Understandability**: would a patient or family member follow it?
- **Suitability**: could it be handed to a patient as written, after edits, or not at all?

A short review form follows each version. Brief notes are very helpful: they tell us exactly what the models get right and wrong.

About the source reports

The reports come from public, de-identified research datasets. Placeholders such as [person], {organization} and {omitted} are part of the original anonymisation, not errors. The discharge summaries are from a Brazilian hospital dataset (SUS is Brazil's public health system); the pathology and radiology reports appear to be English translations of German and Turkish originals, so some wording in the originals is unusual.

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DISCHARGE SUMMARY · DS-01

Original report

201 words. Followed by 2 simplified versions.

male, 42 years old. {organization}. 26/12/2021. stay: 23 h, location: ward, payer: sus. self-referral. emergency. hospital care. [person] with crohn's disease, admitted due to abdominal sepsis associated with acute on chronic kidney disease. [person] with crohn's disease, recently underwent a surgical procedure with ileocolectomy and creation of a new anastomosis after acute perforative abdomen. admitted today to a monitored bed in the red room with hypotension, fever and fainting for 1 day. [person] was recently admitted to [organization] for treatment of intra-abdominal collection, increased urinary output via ileostomy and worsening of renal function, being discharged on 11/15 with a schedule of ertapenem 1 g/day, intramuscularly, for 4 weeks. [person] presents with borderline blood pressure, eupneic on room air. at the time of evaluation, he denies any complaints. he reports maintaining physiological diuresis, without complaints. reports good ostomy function. antibiotic therapy with vancomycin meropenem was started, and [person] was transferred to the intensive care unit of [organization], under the care of the proctology team accepted by [person] from the intensive care unit, {omitted}. meropenem vancomycin replacement of kcl 6% orally continuous medications: loperamide, calcium carbonate, cholestyramine, muciloid. 12/27/2021. follow-up appointment. [person] is discharged on leave to follow-up appointment for hospitalization on 01/22.

Simplified version A

Your 42-year-old male family member was admitted to the hospital on 26/12/2021 for an emergency stay that lasted 23 hours. He was brought in because of abdominal sepsis (a serious infection in the belly area) which was associated with acute on chronic kidney disease (a sudden worsening of long-term kidney problems). He has a history of Crohn's disease (a condition that causes inflammation in the digestive tract) and is post-ileocolectomy status (meaning he has previously had surgery to remove part of the intestine). Recently, he had an acute perforative abdomen (a hole in the intestine) and an intra-abdominal collection (a pocket of fluid or pus in the belly), for which he underwent an ileocolectomy and creation of a new anastomosis (a surgical connection to join the remaining parts of the intestine together).

When he arrived, he had hypotension (low blood pressure), which was borderline compared to the normal range of 120/80 mmHg. He also had a fever, which is higher than the normal body temperature range of 36.5 to 37.5 °C, and he experienced fainting. His kidneys were critically affected, indicated by abnormal creatinine levels, which are normally around 0.6 to 1.2 mg/dL. He was transferred to the Intensive Care Unit (ICU) on 26/12/2021 for close monitoring.

To treat the infection, he was given strong antibiotics: Vancomycin and Meropenem. He is also receiving KCl 6% (a potassium supplement) orally to keep his heart and muscles working properly. For his ongoing management, he is taking several continuous medications: Loperamide to stop diarrhea, Calcium carbonate for bone health and digestion, Cholestyramine to bind bile and help with diarrhea, and Muciloid to soothe the gut lining. Previously, he was prescribed Ertapenem, an antibiotic, at a dosage of 1 g/day given intramuscularly for 4 weeks.

He is currently alive and has been discharged. You must keep him on antibiotic therapy with Vancomycin and Meropenem, continue the potassium replacement, and manage his Crohn's disease medications as directed. Please take him for a follow-up appointment on 12/27/2021. He is also scheduled for hospitalization on 01/22.

Please return to the hospital immediately if you notice any of these warning signs: if his fever returns, if he experiences fainting or dizziness, if his abdominal pain worsens, or if his ostomy output changes significantly.

Review of Version A	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Simplified version B

The patient is a 42-year-old male who was admitted to {organization} on 26/12/2021. He had a hospital stay of 23 hours in a ward, and the payer was sus. He came to the hospital through self-referral as an emergency case for hospital care. The patient has Crohn's disease (a long-term condition that causes inflammation in the digestive tract). He was admitted because of abdominal sepsis (a serious, life-threatening infection in the belly area) associated with acute on chronic kidney disease (a sudden worsening of long-standing kidney problems).

Recently, the patient underwent a surgical procedure involving an ileocolectomy (surgery to remove the last part of the small intestine and the first part of the large intestine) and the creation of a new anastomosis (a surgical connection between two parts of the intestine) after an acute perforative abdomen (a sudden hole in the intestinal wall). He was admitted today to a monitored bed in the red room (a specific area for patients needing close observation) due to hypotension (low blood pressure), fever, and fainting for 1 day.

Previously, the patient was admitted to [organization] for treatment of an intra-abdominal collection (a pocket of fluid or pus inside the belly), increased urinary output via ileostomy (more urine passing through the surgical opening in the belly), and worsening of renal function (kidney function getting worse). He was discharged on 11/15 with a schedule of ertapenem 1 g/day, intramuscularly (injected into the muscle), for 4 weeks. Ertapenem is an antibiotic given to fight bacterial infections.

At the time of evaluation, the patient presented with borderline blood pressure. Normal blood pressure is typically around 120/80 mmHg; his was on the lower side but not critically low. He was eupneic (breathing normally) on room air. He denied any complaints at that time. He reported maintaining physiological diuresis (normal urine output), without complaints. He also reported good ostomy function (the surgical opening in his belly was working well).

Antibiotic therapy with vancomycin and meropenem was started. Vancomycin and meropenem are strong antibiotics given to fight the serious infection. The patient was transferred to the intensive care unit (ICU) of [organization]. The ICU is a special unit for patients who need constant monitoring and life support. He was under the care of the proctology team (doctors who specialize in diseases of the rectum and anus). The ICU team accepted the patient from the intensive care unit, {omitted}.

Medications included meropenem and vancomycin (antibiotics for infection), and replacement of KCl 6% orally (potassium chloride taken by mouth to replace lost potassium, which is important for heart and muscle function). Continuous medications included loperamide (to stop diarrhea), calcium carbonate (a calcium

supplement for bones), cholestyramine (to bind bile acids and reduce diarrhea), and muciloid (a medication to protect the stomach lining).

On 12/27/2021, there was a follow-up appointment. The patient is discharged on leave to follow-up appointment for hospitalization on 01/22.

Review of Version B	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Original report

749 words. Followed by 2 simplified versions.

1. bloodstream infection associated with Schilley catheter in the right femoral vein sac 11/08 2. bacterial endocarditis associated with catheter echocardiogram 11/18: presence of vegetation in the mitral valve 3. pyogenic spondylodiscitis in l3/l4 11/05: low back pain, gait disturbance, computed tomography of the lumbosacral spine 11/12: findings suggestive of a fracture due to bone fragility of the vertebral body of l1, gas foci in the spinal canal at the level of l2l3 that may be related to an extruded disc herniation with degenerative disc disease, and an infectious or inflammatory process cannot be ruled out, lumbar magnetic resonance imaging 25/11/2020: infectious spondylodiscitis with anterior epidural abscess 4. left ileopsoas abscess, magnetic resonance imaging 25/10/2020: intramuscular collection in the ileopsoas left 15 x 2.1 x 1.8 cm with edema and adjacent post-contrast enhancement. admission: 11/08/2020, discharge: 01/02/2021, stay: 01 m 24 d 15 h, location: ward: ward payer: sus. [person] attends hemodialysis session with severe left lower back pain that had started 3 days ago, associated with asthenia, and altered gait. [person] who was accompanying him, denies fever, nausea, and vomiting. Denies other associated symptoms. catheter dressing in the right femoral vein was secreting. [person] evolved during hospitalization with a diagnosis of bloodstream infection by staphylococcus aureus mrsa, complicated by l2l4 spondylodiscitis, mitral valve infective endocarditis, and ileopsoas muscle abscess. Catheter removed on 11/08 and antibiotic therapy guided by vancomycin culture was started on 11/08. Abscess in psoas was not approached due to interposition of bowel loops in an attempt at percutaneous drainage by the {organization} on 12/02. Comparative radiological improvement was observed in axial lesions and conservative treatment was maintained regarding foci of infection identified during hospitalization. [person] remained afebrile with successively negative blood cultures. As a complication associated with immobility, he developed sacral pressure injury, treated with surgical debridement and local measures. Injury care guidelines were provided by the team. He evolved with the need for hemodialysis on Tuesdays, Thursdays and Saturdays due to arteriovenous fistula in the left upper limb, but local stenosis was noted. He is currently on renal replacement therapy. good general condition, pale 1/4, hydrated, acyanotic, anicteric, afebrile to the touch and eupneic in room air. neurology: conscious and oriented in time, space and person. respiratory auscultation: bilateral vesicular breath sounds with discreet bibasilar crackles, no signs of respiratory distress. cardiac auscultation: irregular heart rhythm, with normal sound sounds, no murmurs. abdomen: protuberant, bowel sounds, normotensive, painless on superficial or deep palpation, no signs of peritoneal irritation. warm and well perfused extremities. upper limbs: fistula in the left upper limb with present thrill. lower limbs: presence of discreet edema and anterior tibial ochre dermatitis. nuclear magnetic resonance 25/10/2020: intramuscular collection in the left ileopsoas 15 x 2.1 x 1.8 cm with edema and post-contrast enhancement adjacent cultures: catheter tip 11/08: vancomycin-sensitive staphylococcus aureus blood culture 11/08: staphylococcus aureus sg catheter 11/08 staphylococcus aureus blood culture 11/11: staphylococcus aureus oxa r in 2 samples blood culture 11/21: citrobacter koseri 1 sample staphylococcus aureus 1 sample blood culture 11/24: staphylococcus capitis blood culture 11/28: negative blood culture 12/01: negative blood culture 12/02: negative. attempted percutaneous abscess puncture on 12/02 not performed due to interposition of intestinal loops. nuclear magnetic resonance report of the pelvis performed on 12/16. agreed with [organization] for chronic patients to attend the hemodialysis room at [organization] on Tuesdays, Thursdays and Saturdays, with transportation from the city of origin to the unit carried out by stretcher taxi {omitted}. therefore, we discharge the patient with the following pending issues: observations: administer 0.5 g of vancomycin after each scheduled hemodialysis session antibiotic therapy until 01/09/2021, completing 6 weeks after the last negative blood culture assessment of arteriovenous fistula: phlebography of arteriovenous fistula indicated assess fistulography by brachial to determine the site of stenosis and then perform phlebography or assess pre-procedure with ultrasound in the room. [person] placed on list. [organization] will contact you to schedule a treatment, as recorded in the note by the sacral pressure injury specialist team: [person]'s outpatient follow-up appointment on 16/01/2021 for pressure injury reassessment. Instructions for home care of the injury were provided. Cardiology follow-up for amyloidosis investigation: [person] missed a follow-up appointment appointment on 09/09, scheduled for 01/13 for motor rehabilitation. We are awaiting an appointment by the physiatry department at [organization] requested on 12/27. Check the nuclear magnetic resonance report of the pelvis from 12/16. Previous chronic medications were maintained. Dialysis room Tuesdays, Thursdays, and Saturdays. Amyloidosis cardiology outpatient clinic 01/13. Pressure injury surgery outpatient clinic 01/16.

Simplified version A

The patient was admitted to the hospital on 11/08/2020 and discharged on 01/02/2021, staying for a total of 1 month, 24 days, and 15 hours in the ward. The main reason for admission was a bloodstream infection (an infection in the blood) linked to a Schilley catheter (a type of tube) placed in the right femoral vein (the large vein in the thigh). During the stay, the patient was diagnosed with bacterial endocarditis (an infection of the heart valves), specifically affecting the mitral valve, which was confirmed by an echocardiogram (an ultrasound of the heart) on 11/18 that showed vegetation (a mass of bacteria and cells) on the valve. Additionally, the patient developed pyogenic spondylodiscitis (an infection of the spine bones and the discs between them) at the L3/L4 level, which caused low back pain and difficulty walking. A CT scan (a type of X-ray) of the lower spine on 11/05 and an MRI (a detailed magnetic scan) on 25/11/2020 confirmed this infection and an anterior epidural abscess (a pocket of pus in the space around the spinal cord). There was also a left ileopsoas abscess (a collection of pus in the deep hip muscle), seen on an MRI on 25/10/2020, measuring 15 x 2.1 x 1.8 cm.

The patient initially presented with severe left lower back pain that started three days prior, along with weakness and an unsteady walk. The accompanying person reported no fever, nausea, or vomiting, but the catheter dressing on the right thigh was leaking. The infection was caused by Staphylococcus aureus MRSA (a type of bacteria that is resistant to many antibiotics). The catheter was removed on 11/08, and antibiotic treatment was started the same day, guided by Vancomycin (an antibiotic used to fight this specific bacteria). An attempt to drain the psoas abscess with a needle on 12/02 was not possible because bowel loops (parts of the intestine) were in the way. However, follow-up scans showed improvement in the spine infections, so the team continued with conservative treatment (non-surgical care). The patient remained without fever, and blood cultures (tests to check for bacteria in the blood) became negative, meaning the infection in the blood was cleared.

Due to being bedridden, the patient developed a sacral pressure injury (a bed sore on the tailbone), which was treated with surgical debridement (cleaning of the wound) and local care. The patient required hemodialysis (a machine that cleans the blood when the kidneys cannot) on Tuesdays, Thursdays, and Saturdays using an arteriovenous fistula (a surgically created connection between an artery and vein in the arm for dialysis) in the left upper limb, though a local narrowing (stenosis) was noted. At discharge, the patient was in good general condition, slightly pale, well-hydrated, without fever, and breathing normally on room air. The patient was conscious and oriented. Heart sounds were irregular but normal in quality, with no murmurs. The abdomen was soft and painless. The left arm fistula was working well, and the legs had mild swelling and some skin discoloration on the shins.

The patient is being discharged with several pending issues. First, the patient must take 0.5 g of Vancomycin after each scheduled hemodialysis session until 01/09/2021, which completes the 6-week course required after the last negative blood culture. Second, the arteriovenous fistula needs evaluation; a phlebography (an X-ray test using dye to look at veins) is indicated to find the site of the narrowing, and [person] has been placed on the list for this. [organization] will contact the family to schedule this treatment. Third, regarding the sacral pressure injury, [person] has an outpatient follow-up appointment on 16/01/2021 for reassessment, and instructions for home care of the wound have been provided. Fourth, there is a cardiology follow-up for amyloidosis (a condition where abnormal protein builds up in organs); the patient missed a previous appointment on 09/09, and a new one is scheduled for 01/13. Fifth, for motor rehabilitation, the physiatry department at [organization] requested on 12/27 is awaiting an appointment. Finally, the family should check the MRI report of the pelvis from 12/16. The patient's previous chronic medications have been continued. The dialysis schedule remains Tuesdays, Thursdays, and Saturdays. The cardiology appointment for amyloidosis is on 01/13, and the pressure injury surgery clinic appointment is on 01/16. Transportation for dialysis will be arranged by stretcher taxi {omitted}.

Review of Version A	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Simplified version B

Your son, [person], was admitted to the hospital on 11/08/2020 and discharged on 01/02/2021, staying for a total of 1 month, 24 days, and 15 hours. He came in with severe pain in his left lower back, general weakness (asthenia), difficulty walking (altered gait), and fluid leaking from the dressing of his catheter (a tube used for dialysis) in the right femoral vein. The main problem found was a serious infection in the blood (bloodstream infection) caused by a resistant type of bacteria called Staphylococcus Aureus (MRSA). This infection spread to other parts of his body, causing infective endocarditis (infection of the heart valve), specifically on the mitral valve, where a clump of bacteria called vegetation was found. It also caused pyogenic spondylodiscitis (infection of the spine and the discs between the bones) at the L3/L4 level, with an abscess (a pocket of pus) in front of the spinal cord called an anterior epidural abscess. Additionally, he had a left ileopsoas abscess, which is a collection of pus in the deep hip muscle, measuring 15 x 2.1 x 1.8 cm. He also developed a sacral pressure injury (a bed sore on the tailbone) due to lying in bed for a long time. His kidneys are not working well (Chronic Kidney Disease), so he needs hemodialysis (a machine that cleans the blood) three times a week. There is also a narrowing (stenosis) in his arteriovenous fistula (the special blood vessel connection used for dialysis) in his left arm. Doctors also suspect he has amyloidosis (a condition where abnormal proteins build up in organs).

During his stay, the catheter was removed on 11/08/2020. He started taking an antibiotic called Vancomycin, which is used to kill the resistant bacteria in his blood and body. The dose is 0.5 g given after each scheduled hemodialysis session. He continues this antibiotic therapy until 01/09/2021. Doctors tried to drain the pus from the hip muscle using a needle (percutaneous abscess puncture) on 12/02/2020, but they could not do it because intestines were in the way. Instead, they treated it with antibiotics. The sacral pressure injury was cleaned surgically (surgical debridement) to remove dead tissue. His blood cultures (tests to check for bacteria in the blood) became negative, meaning the infection in the blood is clearing up. His heart rhythm was irregular, and he had some mild crackles (crackling sounds) when breathing, but he was not in distress.

He will continue hemodialysis on Tuesdays, Thursdays, and Saturdays. He needs to see the Amyloidosis Cardiology Outpatient Clinic on 01/13/2021 to check for the suspected protein buildup. He must attend the Pressure Injury Surgery Outpatient Clinic on 01/16/2021 to check the healing of the bed sore. He also needs a Motor Rehabilitation appointment, which the Physiatry department will schedule. Finally, he needs a Phlebography or Ultrasound to check the narrowing in his dialysis fistula; the date for this will be scheduled by the hospital.

Please watch for these warning signs and contact the doctor immediately if you see them: fever or chills; increased pain in the back or hip; redness, swelling, or discharge from the dialysis fistula site; or worsening of the sacral wound, such as a bad smell, increased pain, or pus. For home care of the sacral wound, please follow the instructions given by the team. The hospital stay is over, and he is alive and stable.

Review of Version B	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

DISCHARGE SUMMARY · DS-03

Original report

578 words. Followed by 2 simplified versions.

1sensitive pulmonary tuberculosis. diagnosis 2019, ripe 07/02/2020 ri 07/04 2 pulmonary complications secondary to tuberculosis: spontaneous pneumothorax on the right chest drainage on the right on 03/02/2020 pleurostomy 03/03/2020 closure of pleurostomy lung collapse, pleural drainage, heimlich valve on 10/03 removal of the heimlich valve from the drain, keeping the drain open for ostomy on 27/03 new pleurotomy 31/03 exploratory thoracotomy on an emergency basis 07/04 due to bleeding in tuberculous caverns significant oozing bleeding 07/04 packing of the cavity with compresses postoperatively 11/04/2020 right pneumonectomy by lateral thoracotomy tracheostomy, without complications chest drain and Tracheostomy removed on 06/06. 3 surgical wound infections. Multidrug-resistant Acinetobacter baumannii treated with vancomycin meropenem amikacin antibiotic therapy suspended on 02/04 after clinical improvement and procalcitonin of 0.09 4 hypovolemic shock due to surgical wound bleeding pleurostomy requiring urgent surgical approach 07/04 5 cardiopulmonary arrest 10/04 in asystole / 4 min secondary to hypoxia. admission: 03/02/2020, discharge: 09/05/2020, stay: 03 m 06 d 07 h, ward bed, payer: sus. [person] referred with a persistent cough for 4 weeks, dry, without expectoration, associated with night sweats. He also reports that 3 days ago he had a fever, not measured, associated with dyspnea, and also reports 1 day ago, sudden pain in the right hemithorax, radiating to the right cervical region, associated with significant and progressive worsening of dyspnea. He denies bloody expectoration or sputum. He reports being in prison for approximately 3 years, for 6 years, and has been asymptomatic since then. He also reports weight loss of approximately 8 kg, not quantified, 1 month ago. He denies urinary or intestinal changes. regular general condition, pale 1/4, hydrated, acyanotic, anicteric, conscious and oriented cardiac auscultation: normal rhythmic sounds, normal S1 and S2, no murmurs, heart rate 85 beats per minute respiratory auscultation vesicular breath sounds present in the left hemithorax, abolished in the right hemithorax, with asymmetry of expansion, no adventitious sounds, saturation 98%. abdomen: flat, bowel sounds, flaccid, no peritonism. extremities: no edema, no signs of deep vein thrombosis, maintained peripheral perfusion. surgical wound in good condition and dry, presence of suture without dehiscence. sensitivity test for mycobacterium tuberculosis material: sputum collection: 02/03/2020 available: 28/02/2020 released: 28/02/2020 method: dna hybridization geno type mtbdr plus rifampin element. sensitive isoniazid. sensitive mycobacteria bacilloscopy tuberculosis material: sputum collection: 02/03/2020 available: 02/07/2020 released: 02/07/2020 method: ziehlnelsen element result. positive: 10 to 99 baar were found in 100 observed fields. right chest drainage on 02/03/2020 pleurostomy 03/03/2020 pleurostomy closure lung collapse pleural drainage Heimlich valve on 03/10 removal of the Heimlich valve from the drain, keeping the drain open for ostomy on 03/27 new pleurotomy 03/31 emergency exploratory thoracotomy 04/07 due to bleeding in tuberculous cavernosal bleeding significant oozing 04/07 packing of the cavity with compresses postoperatively 04/11/2020 right pneumonectomy by lateral thoracotomy tracheostomy, uneventful chest drain and tracheostomy removed on 06/06. only exams for follow-up appointment. I continue treatment with RI. I provide 2 months of prescriptions. I continue oral iron replacement. We hold a meeting with [person], together with the thoracic surgery team, and we clarify any doubts regarding the condition and treatment. [person] understands the guidelines and appreciates the care of the team. I request exams for a follow-up appointment. We schedule a follow-up appointment at [organization] and at the thoracic surgery team. We advise [person] of warning signs and instructed to call with concerns. We provide a report and contact [organization] for assisted treatment. Infectious diseases and thoracic surgery. Good general condition, no complaints.

Simplified version A

The patient was diagnosed with sensitive pulmonary tuberculosis (lung infection caused by the TB bacteria) in 2019. He was admitted to the hospital on 03/02/2020 and discharged on 09/05/2020, staying for 3 months, 6 days, and 7 hours in a ward bed. The insurance payer was listed as sus. He was referred to the hospital because he had a persistent dry cough for 4 weeks, without coughing up any phlegm, along with night sweats. Three days before admission, he had a fever (which was not measured) and difficulty breathing. One day before admission, he had sudden pain in the right side of his chest that spread to the right side of his neck, and his breathing difficulty got much worse. He did not cough up any blood. He mentioned that he had been in prison for about 3 years out of the last 6 years and had no symptoms since then. He also reported losing about 8 kg in weight one month ago. He did not have any changes in his urine or bowel movements.

On examination, he looked generally well but was pale (pale skin), hydrated, and not blue in color. He was conscious and aware of his surroundings. His heart sounds were normal, with a heart rate of 85 beats per minute (normal range is 60 to 100 beats per minute, so his was within the normal range). When listening to his lungs, normal breath sounds were heard on the left side, but no sounds were heard on the right side, and the right side of his chest did not expand as much as the left. His oxygen saturation was 98% (normal is 95% or higher, so this was within the normal range). His abdomen was flat and soft, with normal bowel sounds and no signs of infection in the abdominal lining. His legs had no swelling and no signs of blood clots. His surgical wound was dry and in good condition, with stitches intact.

Tests were done to confirm the type of TB. A sputum (phlegm) sample collected on 02/03/2020 was tested using a DNA method called Geno Type MTBDR Plus. The result showed the TB bacteria were sensitive to rifampin and isoniazid, meaning these medicines would work against his infection. Another sputum sample collected on 02/03/2020 was tested using a microscope method called Ziehl-Neelsen. The result was positive, showing 10 to 99 bacteria in 100 fields of view, which confirms the presence of TB.

During his stay, he had several complications related to the TB. First, he had a spontaneous pneumothorax (air leaking into the space around the lung, causing it to collapse) on the right side. A chest drain (a tube placed in the chest to remove air or fluid) was put in on 03/02/2020. A pleurostomy (a procedure to create an opening in the chest wall to drain fluid) was done on 03/03/2020. The pleurostomy was closed, but the lung collapsed again, so a pleural drain and a Heimlich valve (a one-way valve attached to the drain to let air out but not back in) were used on 10/03. The Heimlich valve was removed on 27/03, but the drain was kept open. A new pleurotomy (another chest opening) was done on 31/03. On 07/04, he had an emergency exploratory thoracotomy (a major surgery to open the chest to look inside) because of bleeding from TB cavities (hollow spaces in the lung caused by TB). There was significant oozing bleeding. The cavity was packed with compresses (gauze pads) to stop the bleeding. On 11/04/2020, he had a right pneumonectomy (surgical removal of the entire right lung) through a lateral thoracotomy (side chest incision). A tracheostomy (a small opening made in the windpipe to help with breathing) was also done. This surgery went without complications. The chest drain and tracheostomy tube were removed on 06/06.

He also had surgical wound infections caused by a type of bacteria called Multidrug-resistant Acinetobacter baumannii (a germ that is hard to kill with common antibiotics). He was treated with three antibiotics: Vancomycin (an antibiotic to fight the infection), Meropenem (another strong antibiotic), and Amikacin (an antibiotic given to fight the bacterial infection). The antibiotic treatment was stopped on 02/04 after he improved clinically and his procalcitonin level (a blood test that shows how severe a bacterial infection is) was 0.09 (a low level, indicating the infection was under control).

On 07/04, he had hypovolemic shock (a dangerous drop in blood pressure due to losing too much blood) because of bleeding from the surgical wound and the pleurostomy site. This required urgent surgery. On 10/04, he had a cardiopulmonary arrest (his heart and breathing stopped) for 4 minutes due to asystole (a flatline where the heart does not beat) caused by low oxygen levels (hypoxia). He was resuscitated and recovered.

At discharge, he was in good general condition with no complaints. He was continuing his TB treatment with RI (Rifampin and Isoniazid). He was given 2 months of prescriptions. He was also continuing oral iron replacement (medicine to treat low iron levels). A meeting was held with [person], along with the thoracic surgery team, to explain the condition and treatment. [person] understood the instructions and appreciated the care. Exams were requested for a follow-up appointment. A follow-up was scheduled at [organization] and with the thoracic surgery team. [person] was advised about warning signs and told to call if he had any concerns. A report was provided, and [organization] was contacted for assisted treatment involving infectious diseases and thoracic surgery.

Review of Version A	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Simplified version B

Your main diagnosis is Sensitive Pulmonary Tuberculosis (TB in the lungs), which was first diagnosed in 2019. You also developed several serious complications related to this TB. These include a spontaneous pneumothorax (air leaking into the space around the right lung), lung collapse, a surgical wound infection caused by a tough bacteria called Multidrug-resistant Acinetobacter baumannii, hypovolemic shock (a dangerous drop in blood volume due to bleeding from the surgical site), and a cardiopulmonary arrest (a temporary stop of the heart and breathing) that occurred on 10/04 due to low oxygen levels.

When you were admitted, you had been suffering from a persistent dry cough for 4 weeks, night sweats, and a fever that was not measured. You also experienced shortness of breath (dyspnea) and, one day before admission, sudden pain in your right chest that spread to your neck, along with worsening breathlessness. You had also lost about 8 kg in weight over the previous month. Your general condition was noted as pale 1/4 (mildly pale, which is not the normal skin color), but you were hydrated and conscious. Your heart rate was 85 beats per minute, which is within the normal range of 60 to 100 beats per minute. Your oxygen saturation was 98%, which is within the normal range of 95-100%. However, your respiratory exam showed abolished breath sounds in the right hemithorax (no air movement heard in the right lung), whereas normal breath sounds should be heard in both lungs. A sputum test called Bacilloscopy (Ziehl-Neelsen) came back positive, showing 10 to 99 TB bacteria in 100 fields, while a normal result would be negative (0 bacilli). A blood test called Procalcitonin was 0.09, which is within the normal range of less than 0.5 ng/mL, indicating that the infection was resolving.

You underwent several procedures during your stay. First, a right chest drainage was placed on 03/02/2020 to remove air or fluid. Then, a pleurostomy (a small opening in the chest wall to drain the lung space) was done on 03/03/2020 and closed on the same day. A Heimlich valve (a one-way valve attached to the drain to let air out but not in) was placed on 10/03/2020 and removed on 27/03/2020. A new pleurotomy (another chest opening) was performed on 31/03/2020. On 07/04/2020, you had an emergency exploratory thoracotomy (a major surgery to open the chest) because of significant bleeding from the TB cavities. The cavity was packed with compresses (gauze pads) to stop the bleeding. On 11/04/2020, you had a right pneumonectomy by lateral thoracotomy (surgical removal of the right lung) and a tracheostomy (a small opening in the windpipe to help with breathing). The chest drain and tracheostomy were removed on 06/06/2020.

For your wound infection, you were treated with strong antibiotics: Vancomycin, Meropenem, and Amikacin. These were suspended (stopped) on 02/04/2020 after your condition improved. You are continuing treatment

with RI (Rifampin/Isoniazid), which are medicines to kill the TB bacteria. You are also taking Oral Iron, a supplement to help make blood and fix anemia (pallor).

Your hospital stay lasted 03 m 06 d 07 h. You are currently alive and in good general condition with no complaints. You must continue your TB medication (RI) and oral iron replacement. You have a follow-up appointment at [organization] and with the thoracic surgery team; the date is not specified, but exams have been requested for this visit. Please watch for warning signs such as any new or worsening shortness of breath, chest pain, fever, or signs of infection at the wound site. If you have any concerns regarding your condition, please call immediately.

Review of Version B	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

DISCHARGE SUMMARY · DS-04

Original report

268 words. Followed by 2 simplified versions.

admission: 12/06, discharge: 12/21, stay: 14 d 11 h, location: ward, payer: sus. 1. heart failure profile c dobutamine on 12/07 and 12/08, 2. abscess in the left thigh, osteomyelitis in the pubis, fournier's syndrome, drainage on 12/03 with ultrasound and drain placement on 12/15, treatment with ceftriaxone guided by culture of abscess escherichia coli multi s colonized antibiotic therapy after shock, incision in intensive care unit on 12/17 with abundant purulent secretion, {omitted}. 3. septic shock from cutaneous urinary focus on 12/16 polymyxin d0 12/15 amikacin and vancomycin on 12/17, 4. chronic kidney disease acute dialysis by fistula on 12/16 5. cardiorespiratory arrest with pulseless electrical activity, 10 minutes after intubation report of previous bradycardia, follow-up appointment after 1 minute of cardiopulmonary resuscitation, 6. vt with pulse after cardiorespiratory arrest, effective electrical cardioversion with a shock. [person] presenting with instability in the last half hour, requiring norepinephrine elevation up to 1.5 mcg/kg/min. progresses with bradycardia, QRS widening, and worsening shock. Given the hyperkalemia even after hemodialysis, 2 ampoules of calcium gluconate and 100 ml of bicarbonate were administered, and the respiratory rate was increased on the ventilator in order to reduce pco2 and consequently k. 2 ampoules of atropine were administered without benefit. [person] progressed to asystolic cardiorespiratory arrest, {omitted} in refractory shock. I declare {omitted} dead, due to absence of central pulses, pupils in fixed mydriasis, absence of heart sounds and asytole on the monitor, as well as absence of brainstem reflexes. The body was released. I request contact with the family. I await the death certificate to be completed. Death. Drainage of Fournier's gangrene. Death.

Simplified version A

The patient was admitted to the hospital on 12/06 and stayed for 14 days and 11 hours, with discharge recorded on 12/21. The primary diagnosis was septic shock (a life-threatening drop in blood pressure caused by a severe infection) originating from a cutaneous urinary focus (an infection in the skin and urinary area). The patient also suffered from heart failure (when the heart cannot pump blood effectively), an abscess (a pocket of pus) in the left thigh, osteomyelitis (bone infection) in the pubis (the bone at the front of the pelvis), Fournier's syndrome (a severe, rapidly spreading infection of the genital and perineal area, also known as Fournier's gangrene), and chronic kidney disease (long-term damage to the kidneys).

Upon admission, the patient showed signs of instability, bradycardia (a slower than normal heart rate), QRS widening (an abnormality in the electrical signal of the heart seen on an ECG), and worsening shock. A critical finding was refractory shock, meaning the blood pressure did not respond to treatment, requiring high-dose norepinephrine (a medicine to raise blood pressure) at a level of 1.5 mcg/kg/min, which is significantly higher than the normal typical starting range of 0.01-0.1 mcg/kg/min. The patient also had hyperkalemia (high levels of potassium in the blood), which is dangerous for the heart; the normal range for potassium is 3.5-5.0 mmol/L. Additionally, the patient experienced an asystolic cardiorespiratory arrest (a condition where the heart stops beating and breathing stops, with no electrical activity in the heart), which is a critical deviation from the normal state of regular heartbeat and breathing.

Several procedures were performed during the stay. On 12/07 and 12/08, a heart failure profile with dobutamine (a test to check how well the heart pumps) was conducted. On 12/03, drainage of the abscess in the left thigh was performed. On 12/15, an ultrasound was done and a drain was placed. On 12/16, the patient underwent acute dialysis by fistula (a machine that cleaned the blood because the kidneys could not) and was intubated (a tube was placed in the airway to help with breathing). On the same day, 12/16, cardiopulmonary resuscitation (CPR, chest compressions and breaths to restart the heart) was performed, followed by electrical cardioversion (a shock to the chest to reset the heart rhythm). On 12/17, an incision (a surgical cut) was made in the intensive care unit, revealing abundant purulent (pus-filled) secretion.

The patient was treated with several medications. Ceftriaxone was given as an antibiotic to fight the bacterial infection. Polymyxin, a strong antibiotic for resistant bacteria, was administered on 12/15. Amikacin and

Vancomycin, both antibiotics to fight infection, were given on 12/17. Norepinephrine was used to raise blood pressure, up to 1.5 mcg/kg/min. To protect the heart from the high potassium levels, 2 ampoules of calcium gluconate were administered. Additionally, 100 ml of bicarbonate was given to balance blood acidity and help lower potassium. 2 ampoules of atropine, a medicine to increase heart rate, were also administered.

The patient's condition deteriorated with ventricular tachycardia with pulse (a fast, irregular heart rhythm where the heart still beats) following the cardiorespiratory arrest. Despite treatments including the increase in respiratory rate on the ventilator to reduce carbon dioxide and consequently potassium, the patient progressed to asystolic cardiorespiratory arrest. The patient died due to the absence of central pulses, fixed dilated pupils, absence of heart sounds, asystole on the monitor, and absence of brainstem reflexes. The body was released to the family.

Review of Version A	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Simplified version B

The patient was admitted on 12/06 and passed away on 12/21, staying in the hospital ward for 14 days and 11 hours. The patient had heart failure profile C, which means the heart was not pumping blood effectively, and was given dobutamine, a medicine to help the heart pump stronger, on 12/07 and 12/08. There was an abscess (a pocket of pus) in the left thigh and osteomyelitis (bone infection) in the pubis bone. The patient also had Fournier's syndrome, a severe infection of the skin and tissues around the genitals. On 12/03, the area was drained, and on 12/15, an ultrasound was used to place a drain to remove fluid. The patient was treated with Ceftriaxone, an antibiotic given to fight the bacterial infection, which was chosen based on a culture test that showed Escherichia coli, a type of bacteria that was resistant to many antibiotics. This treatment was started after the patient went into shock, a dangerous drop in blood pressure. On 12/17, a surgical cut was made in the intensive care unit to remove a large amount of pus.

On 12/16, the patient developed septic shock, a life-threatening reaction to the infection spreading from the skin and urinary system. On 12/15, the patient was given Polymyxin, an antibiotic for severe infections. On 12/17, Amikacin and Vancomycin, both antibiotics used to fight different types of bacteria, were administered. The patient had chronic kidney disease, meaning the kidneys were not working well over a long time, and on 12/16, acute dialysis was performed using a fistula (a connection between an artery and vein) to clean the blood because the kidneys could not.

The patient experienced a cardiorespiratory arrest, where the heart and breathing stopped, with pulseless electrical activity, meaning the heart had an electrical signal but no pulse. This happened 10 minutes after

intubation, a procedure where a tube is put into the windpipe to help breathing. There was a report of previous bradycardia, which is a slow heart rate. After 1 minute of cardiopulmonary resuscitation (CPR, chest compressions to restart the heart), the patient had ventricular tachycardia (VT), a fast and irregular heart rhythm, but still had a pulse. An effective electrical cardioversion, a shock given to the chest to reset the heart rhythm, was performed.

In the final hour, the patient became unstable and required Norepinephrine, a medicine to raise blood pressure, up to a high dose of 1.5 mcg/kg/min. The patient's condition worsened with bradycardia (slow heart rate), QRS widening (a sign of heart strain on the monitor), and worsening shock. Because the potassium level in the blood was very high even after hemodialysis, which can stop the heart, 2 ampoules of Calcium Gluconate and 100 ml of Bicarbonate were given to protect the heart and balance the blood. The ventilator settings were increased to lower the carbon dioxide level, which helps reduce potassium. 2 ampoules of Atropine, a medicine to speed up the heart, were given but did not help.

The patient then progressed to asystolic cardiorespiratory arrest, where the heart stopped beating completely. The patient died in refractory shock, meaning the shock could not be reversed. The doctor declared the patient dead due to the absence of central pulses, fixed and dilated pupils (which do not react to light), no heart sounds, and asystole (flat line) on the monitor, as well as the absence of brainstem reflexes. The body was released to the family. The doctor requested contact with the family and is waiting for the death certificate to be completed. The cause of death was recorded as drainage of Fournier's gangrene leading to death.

Review of Version B	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Original report

1370 words. Followed by 1 simplified version.

liver cirrhosis. admission: 09/19/2022, discharge: 10/19/2022, stay: 01 m 11 h, location: ward, payer: SUS. 60 years old, female, from [address], incomplete high school education, works as a cleaner. Anthropometry: estimated weight: 60 kg predicted weight: 57 kg height 164 cm, problem: 1 resolved septic shock with pulmonary focus, acute respiratory failure, hypotension decreased level of consciousness, blood culture with growth of staphylococcus aureus oxas, coronavirus infection in march rtpcr 03/15 2 acute kidney injury kdigo 1 resolved pre-renal, 3 cirrhosis pleural effusion need for paracentesis and relief thoracentesis paracentesis 09/19: blood gas: 1.61 albumin: 0.49 serum albumin: 2.1 100 gb glucose: 164 ldh: 72 serum: 314 thoracentesis 09/19: transudate protein: 2 serum: 7.7 ldh: 72 serum: 279 pH: 7.378 albumin: 0.61 serum: 1.9 amylase: 117 glucose: 183 gb: 100, negative culture thoracentesis 09/23/2022: drained 700 ml. transudate thoracentesis 10/06/2022: drained 1000 ml thoracentesis 10/10/2022: drained 1000 ml. transudate thoracentesis 10/15/2022: drained 1000 ml. history of current illness: [person] diagnosed with liver cirrhosis of etiology to be clarified in follow-up with the gastroclinic at [organization]. underwent a consultation on 02/09/2022, when she was advised to increase the furosemide to 40 mg twice daily and spironolactone to 50 mg/d due to massive ascites, dyspnea, and weight gain. Paracentesis was also performed on the same day, with drainage of 4.1 liters of ascitic fluid. She underwent paracentesis again on 13/09, with 1.4 liters of fluid drained. [person] was referred from the emergency care unit with information from a medical report reporting admission on 18/09/2022, with 85% room air desaturation, obtunded, with peripheral cyanosis and livedo reticularis, followed with uneventful orotracheal intubation. {omitted}, reporting onset of dyspnea this morning. 4 ampoules of furosemide were administered, with no urine output, according to the transport team. I admit [person] intubated, sedated with midazolam and fentanyl, blocked with rocuronium, hemodynamically compensated with norepinephrine 20 ml/h. hospitalization summary: [person] initially admitted to {omitted}, where hypotheses of acute respiratory failure secondary to coronavirus, congestion or sepsis of pulmonary focus were considered. upon admission, empirical ceftriaxone was started and diagnostic paracentesis was performed, with no signs of spontaneous bacterial peritonitis. on the same day, thoracentesis was performed with removal of 1000 ml of fluid, suggestive of transudate, and relief paracentesis was performed with removal of 4000 ml, with albumin being administered due to fluid removal. Also on 09/19, a pocus was performed that showed lines and pleural thickening, which, associated with the history of respiratory failure and need for vasoactive drugs, strengthened the hypothesis of sepsis of pulmonary focus as the cause of respiratory failure. Therefore, it was decided to associate clarithromycin with clindamycin on 09/24, after observing growth of Staphylococcus aureus in blood culture, antibiotic therapy was changed from ceftriaxone clarithromycin to levofloxacin oxacillin as directed by the [organization]. On the same day, it was decided to request an echocardiogram to screen for endocarditis, given the growth of this microorganism in blood culture. On the morning of 09/27, it was decided to escalate antibiotic therapy to meropenem oxacillin due to worsening of CRP associated with increased need for vasoactive drugs. evolved with improvement in the ventilatory status, with extubation on 10/03, and withdrawal of supplemental oxygen therapy on 10/13. resolution of the septic condition, with completion of antibiotic therapy on 10/08. remained hospitalized due to the need for 24-hour nursing care, care for pressure injuries, need for oxygen therapy, management of ascites and pleural effusion, without social support for discharge home. personal history: 1 first diagnosis of liver cirrhosis of etiology to be clarified child b7 meld 9 metabolic?, child b8, meld 12 2021, child b8, meld meld 10 2021 under follow-up since 2019 due to alteration in transaminases biopsy on 07/2020 with grade i fibrosis disease progression? sample not significant? 2021 biopsy: chronic liver disease with intense fibrosis and moderate steatosis, favoring the diagnosis of grade 2 steatohepatitis stage 3 complications: ascites, shp with medium-caliber ve, spontaneous bacterial peritonitis, eh, upper gastrointestinal bleeding etiological investigation: 2020 liver biopsy: widened portal space due to mild grade I fibrosis, mild inflammatory infiltrate with mononuclear predominance grade I, minimal hepatitis at the interface, diffuse macro and microgoticular steatosis plus 60% parenchyma grade III, balloon-shaped degeneration grade I, nuclear vacuolization of hepatocytes mild to moderate steatohepatitis nash 6 therapeutic trial for hai with azathioprine and prednisone 11/03 01/14 no satisfactory response drop in tgp, but maintenance of high tgo hai score before treatment 02/2/2/3/1/2/3 9 fan: 1: 350, anti-smooth muscle 1: 80 alteration of canalicular and transaminases alteration international normalized ratio: 06/08: 1.0 / 29/09: 1.9 / 22/10: 4.4 differential diagnosis: fan: 1: 320 homogeneous nuclear anti-mitochondria non-reactive anti-smooth muscle 1: 80 anti-lkm negative ceruloplasmin: 0.305 alpha feto: 6.17 cytomegalovirus: negative epstein bar negative cea: 4.48 n: 4.9 ca 125: 25 n: 21 ca19.9: 27 shp: digestive endoscopy 2021 medium-caliber esophageal varices with no signs of recent bleeding mild hypertensive gastropathy exanthematous bulboduodenitis 2 weight loss to be clarified 10 kg in 9 months malabsorptive syndrome to be clarified folic acid

deficiency decompensation diabetes mellitus?? occult neoplasia? depressive disorder? investigation: folic acid 1.29 deficient 0.35 3.37 prot/creat ratio 120 300 immunoglobulin iga dosage: 917 40350 vitamin d 04/01: 34 antitga 04/01: positive 3, recent hospitalization: epistaxis, hematuria and melena coagulopathy due to consumption? progression of liver disease? 12/30 01/01 comorbidities allergy to dipyrone chickenpox and measles in childhood tonsillectomy for hypertrophy of pharyngeal tonsils 15 years ago diabetes mellitus 15 years ago systemic arterial hypertension 10 years ago nontoxic multinodular goiter diagnosed in 2019 anxiety disorder, depression diagnosed 6 years ago spondyloarthrosis spine, low back pain painful nodule in the inferolateral quadrant of the right breast with biopsy performed in November 2019, indicating probable fibroadenoma ex-smoker habits claims to have smoked 1 pack/day for only a few months reports a history of drinking 34 cans of beer per weekend, but denies daily use of the substance claims to have stopped 2 years ago denies use of illicit drugs last sexual activity 1 year ago. Denies sexual activity without a condom Denies use of NSAIDs Reports sporadic use of mate tea and lemon balm Denies tattoos and piercings Medications used prior to hospitalization: Insulin NPH 1008 Glifage XR 500 mg: 2 tablets at lunch and 2 tablets at dinner Simvastatin 20 mg/day Losartan 50 mg: 1 tablet every 12 hours Hydrochlorothiazide: 1x day Sertraline 50 mg: 1 tablet every 12 hours Gabapentin 300 mg/day Calcium and vitamin D Amitriptyline 25 mg Propranolol 10 mg every 12 hours Omeprazole 40 mg Furosemide 40 mg twice a day Spironolactone 50 mg/d External exams: 2020 mammogram: Right retroareolar breast nodule, negative biopsy for neoplasia Right breast: BIRADS 4 Left breast: BIRADS 1 upper digestive endoscopy 2020: distal non-erosive esophagitis, enanthematous pangastritis biopsy: helicobacter pylori antibiotic therapy: current: nothing previous ceftriaxone d0 09/19 09/23 clarithromycin d0 09/20 09/23 levofloxacin d0 09/23 09/26 oxacillin d0 09/24 10/08 meropenem d0 09/27 08/10 devices nasoenteral tube valproic acid previous devices: tot 09/18 09/26 tot 09/26 10/03 central venous access right internal jugular vein 09/19 09/28 indwelling urinary catheter 09/18 03/10 cultures: urine culture 09/19: negative urine culture 09/28: negative blood culture 09/19: staphylococcus aureus in 2 samples blood culture 09/24: staphylococcus epidermidis oxa r blood culture 09/28: staphylococcus capitis oxa r pleural fluid 09/19: negative pleural fluid 09/23: negative pleural fluid 09/24: negative ascitic fluid 09/19: negative catheter tip 09/28: negative tests: tomography 09/21: bilateral pulmonary opacities that may represent edema or inflammatory process. bilateral pleural effusion with atelectatic opacities in the lower lobes. cardiomegaly. transthoracic echocardiography: ejection fraction 59% [person], LVEDD 45. marked dilation of the left atrium. preserved ventricular systolic performance. normal left ventricular segmental mobility. mild mitral and tricuspid valve insufficiency. mild left ventricular diastolic dysfunction. schedule: adjustment of diuretic doses, with active monitoring of renal function; performance of relief paracentesis and thoracentesis; progress in weaning from oxygen therapy; assessment of nasoenteric tube removal; if there is good acceptance of oral diet; care for pressure injuries in sacral and trochanteric regions; scheduled return visit at agaf 18/01, having already spoken with [person] and [person]'s family members about the severity of [person]'s situation as well as prognostic possibilities. family members aware of the situation and a shared decision was made not to perform new invasive measures, {omitted}.

Simplified version A

Your mother, a 60-year-old woman, was admitted to the hospital on 19/09/2022 and discharged on 19/10/2022, staying for a total of 1 month and 11 hours. She was admitted due to liver cirrhosis (permanent scarring of the liver), which had led to several serious complications. At the time of admission, she was experiencing septic shock (a life-threatening drop in blood pressure caused by a severe infection), acute respiratory failure (difficulty breathing requiring machine support), and reduced consciousness (obtunded state). Her blood tests showed a blood infection caused by Staphylococcus Aureus bacteria. She also had acute kidney injury (temporary kidney damage), which has since resolved. Other conditions present included pleural effusion (fluid around the lungs), ascites (fluid buildup in the belly), diabetes mellitus (sugar disease), systemic arterial hypertension (high blood pressure), non-toxic multinodular goiter (enlarged thyroid gland), anxiety and depression, spondyloarthrosis (spine joint wear), and a fibroadenoma (a benign breast nodule).

When she first arrived, her oxygen saturation was critically low at 85% (normal range is 95-100%), her blood glucose was high at 164 mg/dL (normal range is 70-100 mg/dL), and her serum albumin was low at 2.1 g/dL (normal range is 3.5-5.0 g/dL). Her INR (a measure of how long it takes blood to clot) was critically high at 4.4 (normal range is 0.8-1.2), indicating a high risk of bleeding. Her LDH (a liver enzyme) was 72 U/L, which is within the normal range of 140-280 U/L, though this value is often monitored in liver disease. Her folic acid level was low at 1.29 ng/mL (normal range is 3.37 ng/mL). Imaging showed bilateral pleural effusion (fluid on

both sides of the lungs) and cardiomegaly (an enlarged heart). She also developed pressure injuries (bedsores) during her stay.

To treat these issues, several procedures were performed. On 18/09/2022, she underwent orotracheal intubation (a tube placed in the windpipe to help her breathe). She was successfully extubated (the tube was removed) on 03/10/2022. On 19/09/2022, a paracentesis (a procedure to drain fluid from the belly) and a thoracentesis (a procedure to drain fluid from around the lungs) were performed. Additional thoracentesis procedures were carried out on 23/09/2022, 06/10/2022, 10/10/2022, and 15/10/2022 to remove accumulated fluid. A transthoracic echocardiography (an ultrasound of the heart) and a POCUS (Point of Care Ultrasound) were also performed on 19/09/2022 to assess her heart and fluid levels.

She received several medications to manage her conditions. For her sugar, she was given Insulin NPH (to control blood sugar) at a dosage noted as 1008, and Glifage XR (Metformin) to control blood sugar, taken as 2 tablets at lunch and 2 tablets at dinner. To lower her cholesterol, she took Simvastatin 20 mg daily. For her high blood pressure, she took Losartan 50 mg, 1 tablet every 12 hours, and Hydrochlorothiazide, a water pill to reduce fluid, once a day. For her mental health, she was prescribed Sertraline 50 mg, 1 tablet every 12 hours, for depression and anxiety. For nerve pain, she took Gabapentin 300 mg daily. For bone health, she received Calcium and Vitamin D. Amitriptyline 25 mg was given for pain or sleep. Propranolol 10 mg every 12 hours was prescribed to protect against bleeding in liver disease. Omeprazole 40 mg was given to reduce stomach acid. To reduce fluid in the belly and lungs, she took Furosemide 40 mg twice a day and Spironolactone 50 mg daily. Antibiotics were used to treat the infection: Levofloxacin was started on 23/09 and stopped on 26/09; Oxacillin was started on 24/09 and stopped on 08/10; and Meropenem, a strong antibiotic for severe infection, was started on 27/09 and stopped on 08/10.

Going forward, her treatment plan includes adjusting the doses of her water pills (diuretics) while closely monitoring her kidney function. She will continue to receive care for her pressure injuries and will gradually wean off oxygen therapy. You must watch for warning signs such as signs of infection (fever, chills), worsening shortness of breath, bleeding (vomiting blood or black stools), confusion or drowsiness, and significant weight gain or swelling. A follow-up appointment with the Gastroenterology department (AGAF) is scheduled for 18/01. The family has been informed about the severity of her condition and the prognosis, and a shared decision was made not to perform new invasive measures.

Review of Version A	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Original report

493 words. Followed by 2 simplified versions.

Material submitted: Prostate.
Clinical data: Known prostate carcinoma.

Pathological-anatomical report:

Macroscopic:
A prostatectomy specimen weighing 50 g and measuring 5.1 x 4.3 x 3.2 cm with bds. dorsally attached seminal vesicles and ductus deferentes. Seminal vesicle on the right measuring up to 3.5 cm, on the left up to 2.5 cm. Ductus deferens on the right up to 6 cm, ductus deferens on the left up to 5.5 cm. Color marking of the prostate ventrally blue and dorsally black.
1st apical margin, right in (A, B), left in (C, D). (4 Bl, kR).
2. base of vesicle, right (A, B), left (C, D) (4 Bl, kR).
3. sedimentation margins of vas deferens, right (A), left (B). (2 Bl, kR).
4. lamellation of the prostate. The cut surface, grayish-beige, firm-elastic, fibrous, partially cystic. Embedding in toto, right half of prostate from apical to basal in (A - U), left half from apical to basal in (V-Z19) (45 Bl, kR).
5. embedding of the seminal vesicles: Transition of the prostate into the right seminal vesicle in (A+B), seminal vesicle right lamellar in (C-F). Transition of the prostate into the left seminal vesicle in (G+H), seminal vesicle lamellated in (I-L) (12 Bl, kR).

Microscopically:
1. fibromuscular tissue with localized prostate glands. No inflammatory infiltrates. In the right-sided area a microfocal area of abnormally positioned glands with partly prominent nucleoli. Basal cell layer not recognizable.
2. fibromuscular tissue with localized prostate glands (more on the left than on the right). No cytomorphologic atypia.
3. wall parts of the ductus deferentes with locotypical structure. No inflammation or atypia.

- b.w. -

4. infiltrates of the tumor described above are visible on both sides of the prostate (more on the right than on the left). In addition to a single-gland structure, confluent growth patterns are also visible. There is an ingrowth into perineural sheaths. The ink-marked margins of the specimen are not reached. No overgrowth of the prostate tissue or infiltration into the seminal vesicle is visible. Residual prostate with an implied nodular structure. No inflammatory infiltrates.
5. seminal vesicles on both sides with locotypical structure. No inflammation or atypia.

Tumor blocks for extra examinations: 4A, 4C, 4Z2.

Assessment:
1st-5th prostatectomy specimen with bilateral infiltrates of acinar adenocarcinoma, perineural sheath infiltration. Tumor-free apical and basal sedimentation lines. Tumor-free seminal vesicles on both sides. Tumor-free ductus deferens deposits on both sides. Residual prostate with adenomyomatosis.

TNM classification (Prostate 2017, 8th edition): pT2c, pN0 (0/12 taking into account parallel submission), L0, V0, Pn1, R0;

Gleason score: 4 + 3 = 7b (60%4 proportion, 40%3 proportion).

Gleason pattern Gleason score ISUP graduation group Applies
= 3 + 3 = 6 1
3 + 4 7a 2
4 + 3 7b 3 X
4 + 4 8 4
4 + 5, 5 + 4, 5 + 5 9-10 5

ICD-O-3 Explanation Applies
8140/3 Adenocarcinoma n.d. X
8020/3 Undifferentiated carcinoma n.a.

We will carry out additional examinations for verification and report back.

Simplified version A

The doctor removed your prostate gland during surgery. The removed tissue weighed 50 grams and measured 5.1 by 4.3 by 3.2 centimeters. Attached to the back of the prostate were the seminal vesicles (small sacs that store fluid for semen) on both sides, as well as the ductus deferentes (the tubes that carry sperm). The right seminal vesicle was up to 3.5 centimeters long, and the left was up to 2.5 centimeters. The right ductus deferens was up to 6 centimeters long, and the left was up to 5.5 centimeters.

The pathologist found a type of cancer called acinar adenocarcinoma (a cancer that starts in the gland cells of an organ) on both sides of the prostate. This cancer is classified with the code 8140/3. The cancer cells were graded using the Gleason score, which is a system from 6 to 10 that shows how abnormal the cells look. Your score is 4 plus 3, which equals 7 (specifically 7b). This places the cancer in ISUP Grade Group 3. This means the cells are moderately aggressive. About 60% of the cancer cells were of the higher-grade type (pattern 4), and 40% were of the lower-grade type (pattern 3).

The report checks if the cancer has spread to nearby structures. It shows perineural invasion (cancer cells growing into the sheaths around nerves), which is present. However, there is no lymphovascular invasion (cancer cells in the lymph vessels or blood vessels). There is no extracapsular extension (cancer growing outside the prostate capsule). There is no seminal vesicle invasion (cancer growing into the seminal vesicles). The seminal vesicles and ductus deferentes on both sides were free of cancer.

The surgeon removed the cancer with clear edges. This is called an R0 resection, meaning the resection margins (the outer cut edge of the tissue removed by the surgeon) are negative. Specifically, the apical margin (the top edge) is negative, the basal margin (the bottom edge) is negative, and the ductus deferens margin is negative. This means no cancer cells were touching the edges of the removed tissue.

In the remaining prostate tissue, the pathologist found adenomyomatosis (a benign condition where the prostate tissue grows in a nodular, thickened way). There were no signs of infection or inflammation.

The pathologist has set aside specific blocks of tissue labeled 4A, 4C, and 4Z2 for extra examinations. Additional examinations for verification are pending, and the doctor will report back with those results.

Review of Version A	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Simplified version B

The tissue examined in this report is the prostate gland, which was removed during surgery. The surgeon also removed the seminal vesicles (small glands that help make semen) and the ductus deferentes (tubes that carry sperm) attached to the back of the prostate. The entire specimen weighed 50 grams and measured 5.1 by 4.3 by 3.2 centimeters. The right seminal vesicle measured up to 3.5 centimeters, and the left one measured up to 2.5 centimeters. The right ductus deferens measured up to 6 centimeters, and the left one measured up to 5.5 centimeters.

Under the microscope, doctors found acinar adenocarcinoma (a type of cancer that starts in the gland cells of the prostate). The cancer was present on both sides of the prostate, with slightly more cancer on the right side than the left. The cancer cells showed a mix of growth patterns, including some single glands and some areas where the cancer cells grew together in clusters. The cancer had grown into the perineural sheaths (the protective coverings around nerves), which is a common way prostate cancer spreads locally. However, the cancer did not grow into the seminal vesicles or the ductus deferentes, and these areas looked normal. The remaining prostate tissue showed a benign condition called adenomyomatosis (a non-cancerous thickening of the prostate tissue).

The report includes a Gleason score of 4 + 3 = 7b. This is a grading system from 6 to 10 that shows how abnormal the cancer cells look under a microscope. A score of 7 means the cancer is moderately aggressive. Specifically, 60% of the cancer cells had a pattern of 4 (more abnormal) and 40% had a pattern of 3 (less abnormal). This places the cancer in ISUP Grade Group 3, which is a middle-level risk category.

The surgical margins, which are the outer cut edges of the tissue removed by the surgeon, were clear. This is called an R0 margin, meaning no cancer cells were touching the edge of the removed tissue. This suggests that the surgeon successfully removed all the visible cancer. The lymph nodes (small bean-shaped structures that filter fluid in the body) were also checked, and no cancer was found in them (pN0). The cancer was confined to the prostate (pT2c) and had not spread to nearby lymph nodes or distant parts of the body.

The report notes that additional tests, including immunohistochemistry (special chemical dye tests that identify specific proteins in cancer cells), will be performed to verify the findings. The final results of these extra tests will be reported separately.

Review of Version B	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Original report

159 words. Followed by 2 simplified versions.

1st supplementary report dated March 14, 2020:

Pathological-anatomical report:

Supplementary immunohistochemical findings (3x IHC):
Ad 4: The tumor strongly expresses AMACR with simultaneous loss of basal cells, detected by MA903 and p63.

Assessment:
Ad 1 - 5. Prostatectomy specimen with bilateral infiltrates of acinar adenocarcinoma, perineural sheath infiltration. Tumor-free apical and basal sedimentation lines. Tumor-free seminal vesicles on both sides. Tumor-free ductus deferens deposits on both sides. Residual prostate with adenomyomatosis.

TNM classification (Prostate 2017, 8th edition): pT2c, pN0 (0/12 taking into account parallel submission), L0, V0, Pn1, R0; Gleason score: 4 + 3 = 7b (60 % proportion of 4s, 40 % proportion of 3s).

Gleason pattern Gleason score ISUP graduation group Applies
= 3 + 3 = 6 1
3 + 4 7a 2
4 + 3 7b 3 X
4 + 4 8 4
4 + 5, 5 + 4, 5 + 5 9-10 5

ICD-O-3 Explanation Applies
8140/3 Adenocarcinoma n.d. X
8020/3 Undifferentiated carcinoma n.d.

Simplified version A

This report is a follow-up to the earlier results from your prostate surgery. The doctors performed special chemical dye tests, known as immunohistochemistry, to look more closely at the cancer cells. These tests showed that the tumor cells strongly produce a protein called AMACR and have lost the normal supporting cells around them, which confirms the diagnosis of cancer.

The main finding is that the removed prostate tissue contains a type of cancer called acinar adenocarcinoma, which is a cancer that starts in the gland cells of the organ. This cancer was found on both sides of the prostate. The cancer cells were also seen growing into the protective coverings of the nerves, a condition called perineural sheath infiltration. However, there is good news regarding the spread of the disease. The surgeon removed the entire prostate, and the outer cut edges of the tissue, called resection margins, were clear. This means no cancer cells were touching the edge of the removed tissue, which is described as R0. Additionally, the cancer had not spread to the lymph nodes (pN0), the lymphatic vessels (L0), or the blood vessels (V0). The seminal vesicles and the ducts that carry sperm were also free of cancer. The remaining healthy prostate tissue showed a benign condition called adenomyomatosis, which is a non-cancerous thickening of the tissue.

The cancer was graded using the Gleason score, a system from 6 to 10 that shows how abnormal the cells look under a microscope. Your score is 7, specifically 4 plus 3. This means the cancer is moderately aggressive. The "4" pattern made up 60% of the cancer, and the "3" pattern made up 40%. In the International Society of Urological Pathology (ISUP) grading system, this places the cancer in Group 3. The final classification is pT2c, which means the cancer was confined to the prostate but involved both sides.

Review of Version A	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Simplified version B

This report is about the prostate (the gland that makes fluid for sperm) that was removed during a surgery called a prostatectomy. The pathologist also examined the bilateral seminal vesicles (the sacs on both sides that store semen) and the bilateral ductus deferens (the tubes on both sides that carry sperm). The main finding is a type of cancer called acinar adenocarcinoma (a cancer that starts in the gland cells of an organ), which was found on both sides of the prostate. This specific cancer type is identified by the code ICD-O-3: 8140/3.

To understand how aggressive the cancer is, doctors use a Gleason score, which is a grading system from 6 to 10 that shows how abnormal the cells look under a microscope. Your score is 4 + 3 = 7 (specifically 7b), which places the cancer in ISUP Grade Group 3. This means the cells are moderately aggressive. The report notes that 60% of the cancer cells are of the higher-grade pattern (pattern 4), while the remaining 40% are of the lower-grade pattern (pattern 3).

The report checks if the cancer has spread to nearby structures. It shows perineural invasion (cancer cells growing along the nerves), which is present. However, there is no lymphovascular invasion (cancer cells in the lymph or blood vessels), no extracapsular extension (cancer growing outside the prostate capsule), and no seminal vesicle invasion (cancer growing into the seminal vesicles). The seminal vesicles on both sides were free of tumor. Additionally, the tumor-free ductus deferens deposits on both sides were examined and found to be clear of cancer.

The surgeon removed the tissue with clear edges, known as resection margins (the outer cut edge of the tissue removed by the surgeon; clear/R0 margins mean no cancer cells were touching the edge). The status is R0, meaning the cancer was completely removed. Specifically, the apical margin (the top edge) was negative (no cancer), and the basal margin (the bottom edge) was negative (no cancer).

To confirm the diagnosis, the lab used immunohistochemistry (special chemical dye tests that identify specific proteins in cancer cells). The results showed that AMACR was strongly positive, which confirms the invasive malignant phenotype (the aggressive nature of the cancer). The tests for MA903 and p63 showed a loss of basal cells (the normal support cells that are usually present in healthy tissue), which also confirms the invasive malignant phenotype. Finally, the remaining healthy prostate tissue showed a condition called adenomyomatosis (a benign thickening of the gland and muscle layers), but this is not cancer.

Review of Version B	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Original report

435 words. Followed by 2 simplified versions.

Material submitted: Prostate.
Clinical data: High risk prostate carcinoma.

Pathological-anatomical report:

Macroscopic:
A prostatectomy specimen weighing 48 g and measuring 6.5 x 4.1 x 3.5 cm with bds. dorsally attached seminal vesicles and ductus deferentes. Color marking of the prostate ventrally blue, dorsally black.
1st (apical margin) Right in (A, B), left in (C, D) (4 Bl, kR).
2. (Bladder base) Right (A, B), left (C, D) (4 Bl, kR).
3. (sedimentation margins of vas deferens) Right (A), left (B) (2 Bl, kR).
4th (prostate) lamellae: The cut surface, grayish-beige, firm-elastic, fibrous, partially cystic. Embedding in toto, right half of prostate from apical to basal in (A - Q), left half from apical to basal in (R- Z7) (33 Bl, kR).
5th (seminal vesicles) transition right prostate seminal vesicle in (A,B), the right seminal vesicle in (C - E). Transition left prostate seminal vesicle in (F+G), and left seminal vesicle in (H+I) (9 Bl, kR)

Microscopic:
1. fibromuscular tissue with localized prostate glands. Abnormally located glandular complexes with prominent nucleoli of the nuclei and not clearly recognizable basal cells.
2. fibromuscular tissue with focal urothelial overlay and here also lymphocyte-dominated inflammatory infiltrates. No atypia.
3. wall parts of the ductus deferentes with locotypical structure. No inflammation or atypia.
4. infiltrates of a gland-forming tumor predominantly with a confluent growth pattern on both sides of the prostate. Cribriform parts are also visible in small foci, as well as solid formations in just under 5 %. There is an ingrowth into perineural sheaths. On the right side, the tumor can also be traced focally over the prostatic capsule into the adjacent fatty/connective tissue.
- b.w. -

5. seminal vesicles on both sides with locotypical structure. No inflammation or atypia.

Tumor blocks for extra examinations: 4B, 4G, 4S, 4W.

Assessment:
1st - 5th prostatectomy specimen with bds. Infiltrates of an acinar adenocarcinoma as well as perineural sheath infiltration and focal extraprostatic growth on the right side. Tumor-free apical and basal sedimentation lines. Tumor-free ductus deferens deposition. Tumor-free seminal vesicles on both sides.

TNM classification (Prostate 2017, 8th edition): pT3a, L0, V0, Pn1, R0; Gleason score: 4 + 3 = 7b (60% proportion of 4s, 35% proportion of 3s); tertiary grade: 5 (5%).

Gleason pattern Gleason score ISUP graduation group Applies
= 3 + 3 = 6 1
3 + 4 7a 2
4 + 3 7b 3 x
4 + 4 8 4
4 + 5, 5 + 4, 5 + 5 9-10 5

ICD-O-3 Explanation Applies
8140/3 Adenocarcinoma n.d. x
8020/3 Undifferentiated carcinoma n.a.

We will carry out additional examinations for verification and report back.

Simplified version A

The doctor examined the prostate gland, which is a small organ in men that helps make semen. This tissue was removed during a major surgery called a radical prostatectomy. The removed prostate weighed 48 grams and measured 6.5 by 4.1 by 3.5 centimeters. Attached to the back of the prostate were the seminal vesicles (glands that make fluid for semen) and the ductus deferentes (tubes that carry sperm), on both the left and right sides.

The microscopic examination found a type of cancer called acinar adenocarcinoma (a cancer that starts in the gland cells of an organ). This cancer was present on both sides of the prostate. The cancer cells were graded using the Gleason score, a system from 6 to 10 that shows how abnormal the cells look. The score was 4 plus 3, which equals 7 (specifically 7b). This places the cancer in ISUP Grade Group 3, which means the cells are moderately aggressive. About 60% of the tumor was made up of the more aggressive pattern 4 cells. There was also a small amount, 5%, of a very aggressive pattern called tertiary grade 5. The cancer had grown into the nerves around the prostate, a finding known as perineural invasion. It had also grown slightly outside the outer layer of the prostate on the right side, called extracapsular extension. However, the cancer had not spread into the blood or lymph vessels (lymphovascular invasion was absent), and it had not reached the seminal vesicles (seminal vesicle invasion was absent).

The surgeon removed all the visible cancer. This is called an R0 resection, meaning the resection margins (the outer cut edge of the tissue removed by the surgeon) were clear. Specifically, the apical margin (the bottom edge), the basal margin (the top edge near the bladder), and the ductus deferens margin (the edge of the sperm tubes) were all negative, meaning no cancer cells were touching these edges. The cancer code for this type of tumor is ICD-O-3 8140/3. The doctors plan to perform additional examinations for verification and will report the results back to you.

Review of Version A	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Simplified version B

The doctor examined a piece of tissue from the prostate gland that was removed during surgery. The prostate is a small gland located below the bladder in men. The removed tissue weighed 48 grams and measured 6.5 by 4.1 by 3.5 centimeters. It also included the seminal vesicles (glands that make fluid for sperm) and the ductus deferentes (tubes that carry sperm). The surgeon marked different parts of the tissue with blue and black ink to help the pathologist identify exactly where each piece came from.

Under the microscope, the pathologist found cancer in the prostate. The type of cancer is called acinar adenocarcinoma, which is a cancer that starts in the gland cells of the prostate. The cancer cells looked quite abnormal. They had large, dark spots inside their nuclei (the control centers of the cells) and were missing a specific layer of healthy cells called basal cells. The cancer grew in a pattern where the cells stuck together in large, connected clusters. In a small area, less than 5 percent of the tumor, the cells grew in solid sheets. The cancer also grew into the sheaths (protective coverings) of the nerves near the prostate. On the right side of the prostate, the cancer had grown slightly past the outer wall of the prostate into the nearby fatty tissue.

The good news is that the cancer did not spread to certain other areas. The pathologist checked the edges of the tissue where the surgeon cut it, known as resection margins. These margins were clear, meaning no cancer cells were touching the cut edges. This is called an R0 margin. The cancer was not found in the apical or basal margins, the ductus deferentes, or the seminal vesicles on either side. The rest of the tissue, including the bladder base area, looked normal with only some mild inflammation.

The cancer is graded using a system called the Gleason score, which ranges from 6 to 10 and shows how abnormal the cells look. A higher number means the cells look more different from healthy cells and may grow faster. This tumor has a Gleason score of 7, specifically 4 plus 3. This means the most common pattern of abnormal cells is pattern 4, which is moderately aggressive, and the second most common is pattern 3, which is less aggressive. The pathologist also noted a small amount, about 5 percent, of very abnormal cells called tertiary grade 5. Overall, this is classified as ISUP Grade Group 3, which indicates a moderately aggressive cancer. The cancer is considered localized to the prostate with slight extension into nearby tissue, but it has not spread to lymph nodes or distant parts of the body. The lab will perform additional tests to confirm these findings and will report the results later.

Review of Version B	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Original report

382 words. Followed by 2 simplified versions.

Material submitted: Prostate and seminal vesicles.
Clinical information: Dignity? TNM? R0?

Pathological-anatomical report:

Sent for frozen section examination:
Macroscopic:
Prostatectomy specimen with seminal vesicles, 48.2 g, 5.5 x 4.7 x 2.9 cm, seminal vesicles up to 3 cm, color marking ventral black, dorsal blue. Deposition margins for frozen section: apical right (A), left (B), basal right (C,D), vascular nerve bundle right (E), left (F)(6 Bl, 6 x SS, R after kerosene embedding).
2nd seminal vesicle deposition margin right (A), left (B). (2 Bl, R)
3. right half of prostate from apex to base (A-L), left half of prostate from apex to base (M-X). (24Bl,kR)
4th (seminal vesicles): Transition prostate right seminal vesicle (A,B), right seminal vesicle (C-E), transition prostate left seminal vesicle (F,G), left seminal vesicle (H-J).

Frozen section diagnosis:
Resection margins of the prostate free of tumor.

Microscopic:
Fibromuscular prostatic parenchyma with extensive infiltration by a predominantly cribriform and glandular-confluent malignant epithelial tumor tissue. The glands are lined by cells with a nuclear-plasma relation shifted in favor of the nucleus, the nuclei somewhat rounded with prominent nucleoli. Partial formation of cleavage artifacts to the adjacent parenchyma.
- b.w. -

The tumor is predominantly located in the right half of the prostate, on the left near the apex, but a small area with small glandular tumor infiltration (< 5%). Seminal vesicles and ductus deferentes with regular structure. Perineural sheath infiltration is not recognizable.

Tumor blocks for extra examinations: 3B, 3D, 3F

Assessment:
1st-4th prostatectomy specimen Infiltration of both prostate lobes (right > left) by acinar adenocarcinoma of the prostate, confined to the prostate. Gleason score 4 + 4 = 8, tertiary grade 3.

Tumor-free apical and basal resection margins as well as tumor-free seminal vesicles, ductus deferentes and vascular nerve bundles on both sides.

Taking into account the parallel submissions, this results in the following

TNM classification (Prostate 2017, 8th edition): pT2b, pN0 (0/19), L0, V0, Pn0, R0;

Gleason score: 4 + 4 = 8; tertiary grade: 3 (5%)

Gleason pattern Gleason score ISUP grading group Applies
= 3 + 3 = 6 1
3 + 4 7a 2
4 + 3 7b 3
4 + 4 8 4 x
4 + 5, 5 + 4, 5 + 5 9-10 5

ICD-O-3 Explanation Applies
8140/3 Adenocarcinoma n.d. x
8020/3 Undifferentiated carcinoma n.a.

Simplified version A

The doctor removed your prostate gland, along with the seminal vesicles (the small sacs that make fluid for semen), the ductus deferentes (the tubes that carry sperm), and the vascular nerve bundles (the bundles of blood vessels and nerves that help with erections). The total weight of the removed tissue was 48.2 grams, and it measured 5.5 x 4.7 x 2.9 cm. During the surgery, a quick test called a frozen section was done, and the result showed that the resection margins (the outer cut edges of the tissue) were free of tumor.

The final detailed examination found acinar adenocarcinoma (a cancer that starts in the gland cells of an organ). This cancer was found in both the right and left sides of the prostate, with more cancer on the right side than the left. The cancer is classified with an ICD-O-3 code of 8140/3. The severity of the cancer is measured by a Gleason score of 4 + 4 = 8, which places it in ISUP Grade Group 4. This score is a grading system from 6 to 10 that shows how abnormal the cells look; a score of 8 means the cells are quite aggressive. A small portion, specifically 5% of the tumor, had a tertiary grade 3 pattern, which refers to a specific type of cell structure.

The good news is that the cancer did not spread outside the prostate. There was no perineural invasion (cancer growing along the nerves), no lymphovascular invasion (cancer in the lymph or blood vessels), no extracapsular extension (cancer breaking through the outer layer of the prostate), and no seminal vesicle invasion (cancer in the seminal vesicles). The resection status is R0, which means the surgeon removed all the visible cancer, and no cancer cells were touching the cut edges. Specifically, the margins at the apical (top) end, the basal (bottom) end, the seminal vesicles, the ductus deferentes, and the vascular nerve bundles were all negative, meaning they were clear of cancer.

The final stage of the cancer is pT2b, which means the tumor was confined to the prostate but involved more than half of it. The pN0 (0/19) result means that out of 19 lymph nodes checked, none had cancer. The L0, V0, and Pn0 codes confirm that there was no cancer in the lymphatics, blood vessels, or nerves. Overall, the cancer was localized to the prostate and was completely removed.

Review of Version A	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Simplified version B

The doctor examined the prostate gland and the seminal vesicles (the small sacs behind the prostate that help make semen) that were removed during surgery. The total weight of the tissue was 48.2 grams, and the main part measured 5.5 by 4.7 by 2.9 centimeters. The seminal vesicles were up to 3 centimeters long. The surgeon marked different parts of the tissue with colors to help the pathologist check the edges carefully.

Under the microscope, the pathologist found a type of cancer called acinar adenocarcinoma (a cancer that starts in the gland cells of the prostate). The cancer cells looked abnormal, with large nuclei (the control centers of the cells) and prominent nucleoli (small structures inside the nuclei). The cancer was mostly in the right half of the prostate, with a smaller amount near the tip on the left side. The cancer did not spread to the seminal vesicles or the ductus deferentes (the tubes that carry sperm). It also did not grow into the nerves or blood vessels.

The cancer is graded using a system called the Gleason score, which ranges from 6 to 10 and shows how abnormal the cells look. This tumor has a Gleason score of 4 plus 4, which equals 8. This means the cells are moderately to highly aggressive. It is also classified as ISUP grading group 4, which is a standard way to group prostate cancers based on how fast they might grow.

The pathologist checked the resection margins (the outer cut edges of the tissue removed by the surgeon). The margins were clear, meaning no cancer cells were touching the edge of the removed tissue. This is called an R0 resection. The cancer was confined to the prostate gland itself and did not spread to nearby lymph nodes (small filters in the body that help fight infection). The final stage is pT2b, which means the cancer was in both halves of the prostate but stayed inside the gland. There was no evidence of spread to lymph nodes (pN0), lymphatic vessels (L0), blood vessels (V0), or nerves (Pn0).

Review of Version B	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Original report

255 words. Followed by 2 simplified versions.

1st supplementary report dated 28.10.2020:

Pathological-anatomical report:

In the meantime, we have carried out immunohistochemical examinations (5 x IHC).

CK AE1/3 marks the tumor cells described above. The tumor infiltrates show a focal overexpression of AMACR. No expression of MA903 or p63. The immunohistochemical staining was not performed in the border area for technical reasons. The tumor infiltrates are border-forming over a distance of 19 mm basally.

Assessment:
Prostatectomy specimen with infiltration of both prostate lobes by an acinar adenocarcinoma of the prostate, Gleason score 4 + 3 = 7b (tertiary grade 5). Infiltration of the seminal vesicles on both sides as well as extraprostatic infiltration.

Tumor-free apical margin and tumor-free ductus deferentes on both sides.

The tumor infiltrates are dorsally and ventrally border-forming in the area of the basal deposit margin on both sides.

Taking into account the parallel submission, the following TNM classification (Prostate 2017, 8th edition) results: pT3b, pN0(0/9), L0, V0, Pn1, R1;

Gleason score: 4 + 3 = 7b (70 % proportion of 4s, 25 % proportion of 3s); tertiary grade: 5 (5 %)

- b.w. -

Gleason pattern Gleason score ISUP graduation group Applies
= 3 + 3 = 6 1
3 + 4 7a 2
4 + 3 7b 3 x
4 + 4 8 4
4 + 5, 5 + 4, 5 + 5 9-10 5

R1 situation:
Infiltrated resection margin Gleason pattern Extent of margin infiltration (mm)
basal (ventral and dorsal) right and left 4 19

ICD-O-3 Explanation Applies
8140/3 Adenocarcinoma n.d. x
8020/3 Undifferentiated carcinoma n.a.

Simplified version A

The doctor removed your prostate gland during surgery, along with the seminal vesicles (the glands that make fluid for semen) and the ductus deferentes (the tubes that carry sperm) on both sides of your body. The lab examined this tissue and found a type of cancer called acinar adenocarcinoma (a cancer that starts in the gland cells of an organ) in both sides of the prostate. This specific type of cancer is identified by the code 8140/3.

To understand how aggressive the cancer is, the pathologist used a grading system called the Gleason score, which ranges from 6 to 10. Your score is 4 + 3 = 7, specifically labeled as 7b. This places your cancer in ISUP Grade Group 3, which means the cells are moderately aggressive. The report breaks down the cell types: 70% of the cancer cells are the more aggressive pattern 4 type, 25% are the less aggressive pattern 3 type, and a small portion, 5%, is a very aggressive pattern 5 type known as tertiary grade 5.

The report also shows that the cancer has spread beyond the main prostate gland. It has grown into the tissue outside the prostate, a condition called extracapsular extension. It has also invaded the seminal vesicles on both sides. Additionally, the cancer cells were found traveling along the nerves, which is called perineural

invasion. However, the cancer was not found in the lymph vessels, so lymphovascular invasion is absent.

The surgeon checked the edges of the removed tissue, known as resection margins, to see if any cancer cells were left behind. A clear margin means no cancer was touching the cut edge. In your case, the top edge (apical margin) was clear, and the edges of the ductus deferentes on both sides were also clear. However, the bottom edge (basal margin) on both the right and left sides, including the front (ventral) and back (dorsal) parts, had cancer cells touching the cut edge. This area of cancer at the edge measured 19 mm. Because cancer was found at the edge, the surgery is classified as R1, meaning some microscopic cancer cells may remain in the body.

To confirm the diagnosis, the lab used immunohistochemistry (special chemical dye tests that identify specific proteins in cancer cells). The test for CK AE1/3 was positive, which marks the tumor cells. The test for AMACR showed focal overexpression, which supports the diagnosis of adenocarcinoma. The tests for MA903 and p63 were negative, meaning those proteins were not expressed. The lab noted that they could not perform the staining in the border area for technical reasons.

Finally, the overall stage of the cancer is described using the TNM classification system. Your stage is pT3b, which means the cancer has spread to the seminal vesicles. pN0(0/9) means that out of 9 lymph nodes checked, none had cancer. L0 and V0 mean no cancer was found in lymph or blood vessels. Pn1 indicates the presence of perineural invasion. The final resection status is R1, confirming that cancer cells were present at the surgical margin.

Review of Version A	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Simplified version B

This report is an update to the results from the surgery where your prostate gland was removed. The doctors performed special chemical dye tests called immunohistochemistry to look for specific proteins in the cancer cells. These tests confirmed the type of cancer found. The cancer cells showed signs of a protein called AMACR, which is often seen in prostate cancer, but they did not show signs of other proteins like MA903 or p63. Because of technical issues, the doctors could not perform these specific dye tests on the very edge of the tissue, but they did examine the edges under a microscope.

The main finding is that the prostate tissue contains a type of cancer called acinar adenocarcinoma, which is a cancer that starts in the gland cells of the organ. This cancer has spread into both the left and right sides of the prostate. It has also grown outside the prostate gland itself and into the seminal vesicles, which are small sacs near the prostate that store semen. The cancer cells look moderately abnormal under the microscope. This is

measured by a Gleason score of 4 plus 3, which equals 7b. The Gleason score is a grading system from 6 to 10 that shows how abnormal the cells look; a score of 7b means the cancer is moderately aggressive. Specifically, 70% of the cancer cells are of the more aggressive type (pattern 4), 25% are of the less aggressive type (pattern 3), and a small 5% portion is of the most aggressive type (pattern 5).

The report also describes the edges of the tissue that the surgeon cut out, known as resection margins. A clear margin means no cancer cells were touching the cut edge. In your case, the top edge (apical margin) and the tubes that carry sperm (ductus deferentes) on both sides were free of cancer. However, cancer cells were found touching the bottom edge (basal margin) on both the front and back sides. This area of cancer at the edge measured 19 millimeters in length. Because cancer cells were found at this cut edge, the result is classified as R1, which means there may be a small amount of cancer left behind in that specific area.

Based on these findings, the cancer is classified as stage pT3b. This means the cancer has grown outside the prostate and into the seminal vesicles. The report notes that no cancer was found in the nearby lymph nodes (pN0), and there was no cancer in the lymphatic vessels (L0) or blood vessels (V0). However, the cancer did involve the nerve tissue (Pn1). This detailed classification helps your medical team understand exactly how far the cancer has spread so they can plan the next steps in your care.

Review of Version B	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

RADIOLOGY REPORT (MRI) · RAD-01

Original report

113 words. Followed by 2 simplified versions.

BRAIN MRI (CRANIAL) [date_1]:
Clinical: Dizziness.
Technique: Three-plane T1-weighted, FLAIR; axial T2-weighted, SWI, and DWI.
Findings:
The brainstem and cerebellum, fourth ventricle, and posterior fossa cisterns are normal.
The bilateral basal ganglia, internal capsules, and corpus callosum are normal.
Except for physiologic cystic perivascular enlargement in the left parietal region, the cerebral white and gray matter is normal.
No abnormality is noted on DWI and SWI images.
The third and lateral ventricles and sulci are normal.
The major cerebral vascular structures are patent and normal.
The paranasal sinuses and middle ear aeration, nasopharynx, and orbits are normal.
No abnormality is noted in the visualized bones.
Impression:
• Cranial MRI findings within normal limits.

Simplified version A

You had a cranial MRI (a detailed brain scan using magnetic fields and radio waves, without any radiation) to help investigate your dizziness. The scan used several specific types of images, including T1-weighted, FLAIR, T2-weighted, SWI, and DWI sequences, and it was performed without using any contrast dye. The good news is that the radiologist's overall impression is that the cranial MRI findings are within normal limits.

Most of the important parts of your head and brain look healthy. The brainstem (the part of the brain that connects to the spinal cord and controls basic life functions) and the cerebellum (the part of the brain that controls balance and coordination) are normal. The fourth ventricle (a fluid-filled space in the brain) and the posterior fossa cisterns (small fluid-filled spaces at the back of the brain) are also normal. The bilateral basal ganglia (deep brain structures involved in movement), the internal capsules (nerve pathways in the brain), and the corpus callosum (the bridge connecting the two halves of the brain) are all normal. The third and lateral ventricles (fluid-filled spaces in the center of the brain) and the sulci (the grooves on the surface of the brain) are normal. The major cerebral vascular structures (the main blood vessels in the brain) are open and normal. The paranasal sinuses (air-filled spaces around the nose and eyes), middle ear aeration (air in the middle ear), nasopharynx (the upper part of the throat behind the nose), and orbits (the eye sockets) are all normal. There are no abnormalities in the visualized bones.

There is one small, non-serious finding: physiologic cystic perivascular enlargement in the left parietal region. This means there are tiny, fluid-filled spaces around blood vessels in the left parietal lobe (the part of the brain involved in sensation and spatial awareness). This is considered a normal, age-related change and is not a sign of disease or damage. Except for this minor, normal variation, the cerebral white and gray matter (the different types of brain tissue) is normal. The DWI and SWI images, which look for specific types of tissue changes or bleeding, showed no abnormalities. No further tests or follow-up actions are recommended based on this report.

Review of Version A	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Simplified version B

This report is from a brain MRI, which is a detailed scan of the head that uses strong magnets and radio waves to create clear pictures of the brain. This type of scan does not use any radiation. The scan was performed to look for the cause of your dizziness.

The good news is that the scan results are completely normal. The radiologist checked many important areas, and everything looks healthy. The brainstem (the part of the brain that connects to the spinal cord and controls basic functions like breathing and heart rate) and the cerebellum (the part of the brain that controls balance and coordination) are normal. The fourth ventricle (a fluid-filled space in the brain) and the spaces around the back of the brain are also normal. The deep parts of the brain, including the basal ganglia (areas involved in movement control), internal capsules (nerve pathways), and corpus callosum (the bridge connecting the two halves of the brain), all appear normal as well.

The brain tissue itself, known as white and gray matter, is normal. There is a tiny, natural widening of the spaces around blood vessels in the left parietal region (the upper middle part of the brain), but this is a normal, harmless variation called "physiologic cystic perivascular enlargement" and is not a disease or a problem. The fluid spaces in the center of the brain, called the third and lateral ventricles, and the grooves on the brain's surface, called sulci, are also normal. The major blood vessels in the brain are open and working correctly.

The scan also checked the areas around the brain, including the paranasal sinuses (air-filled spaces in the face bones), the middle ears, the nasopharynx (the upper part of the throat behind the nose), and the eye sockets, all of which are normal. The bones of the skull that were visible in the scan also show no problems. Special images that look for new strokes or bleeding (DWI and SWI) were also normal. In summary, the brain MRI findings are within normal limits, meaning no abnormalities were found.

Review of Version B	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Original report

329 words. Followed by 2 simplified versions.

BRAIN (CRANIAL) AND TEMPORAL MRI
Clinical information: Left face and jaw pain; peripheral facial paralysis one month ago.
Technique:
Cranial: 3-plane T1-weighted, FLAIR; axial T2-weighted, SWI, and DWI. T1-weighted after IV contrast agent.
Temporal: Axial and coronal T2-weighted, T1-weighted; CISS and DWI. T1-weighted after IV contrast agent.

Findings:
Cranial MRI:
Brainstem, cerebellum, and cerebral parenchyma; ventricles and sulci are normal.
Intracranial DWI, SWI, and post-contrast sections show no specific findings.
In the bone lateral to the left jugular foramen; there is a non-enhancing bone lesion, minimally expansile, located medial to the vertical segment of the left facial nerve. Present also in the MRI examination of the patient's e-Nabız system, and sizes are stable.
The postganglionic segment of the left facial nerve is normal despite close proximity. Residual contrast enhancement in the preganglionic segment is in favor of Bell's palsy, and the cisternal segment of the facial nerve is normal.
In the left portion of the skull base, there are bilateral parietal bone lesions with similar features to the bone lesion described above (FLAIR axial 134-150). These lesions belong to the bone marrow, and periosteal reaction belonging to the outer table accompanies 2 bone lesions at the vertex (FLAIR axial 145 and 157).
Other bone structures are normal.
Temporal bone MRI shows bilateral middle ear and mastoid cell aeration, inner ear structures are normal.

Impression:
• Non-enhancing solid bone lesion in the bone between the left jugular foramen and facial canal; bilateral parietal bone lesions with similar features.
o Temporal bone CT examination is recommended to determine the status of bone trabeculation.
o The lack of contrast enhancement partially excludes neoplastic processes such as glomus jugulare and lymphoma, but should be evaluated in terms of malignant diseases causing bone marrow involvement.
• The temporal segment of the facial nerve is in close proximity but there is no contrast enhancement-thickening. There is mild contrast enhancement in the left preganglionic facial nerve compatible with Bell's palsy.
• No intracranial pathology was detected.

Simplified version A

You had a detailed brain and ear scan called an MRI (Magnetic Resonance Imaging), which uses strong magnets and radio waves to take pictures of your body without using any radiation. This specific scan looked at your brain and the bones around your ears and face. To get clearer pictures, a special dye called contrast was given through an IV. The scan used several different types of image settings, including T1-weighted, FLAIR, T2-weighted, SWI, DWI, and CISS sequences, to look at different parts of the tissue in detail. The reason for this test was to check for the cause of pain in your left face and jaw, as well as the weakness in your left facial muscles that started one month ago.

First, it is good news that many important parts of your head are healthy. The brainstem (the part of the brain that connects to the spinal cord and controls basic life functions), the cerebellum (the part of the brain that controls balance and coordination), and the main brain tissue (cerebral parenchyma) all look normal. The fluid-filled spaces in the brain called ventricles and the grooves on the surface of the brain called sulci are also normal. The special image sequences (DWI, SWI, and post-contrast sections) inside the skull did not show any specific problems. The part of the left facial nerve that comes out of the brain (the postganglionic segment) is normal, as is the section of the nerve in the fluid-filled space around the brain (the cisternal segment). The bones of your skull, other than the specific spots mentioned below, are normal. In the ear area, both middle ears and the air-filled spaces in the mastoid bone (the bony area behind the ears) are clear, and the inner ear

structures are normal.

There are a few areas that need attention. In the bone next to the opening where the jugular vein enters the skull (the left jugular foramen) and near the vertical part of the left facial nerve, there is a small, solid spot in the bone. This spot does not take up the contrast dye, which is a good sign. It is slightly expanded, meaning it has pushed the bone out a tiny bit, but it is the same size as it was in your previous MRI records from the e-Nabız system, so it is stable and likely an old, chronic finding. Because it does not light up with the dye, it makes it less likely to be certain types of tumors like glomus jugulare or lymphoma. However, doctors still need to check if this is related to any serious disease that affects the bone marrow.

Additionally, there are similar spots in the bone marrow of the parietal bones (the bones on the top and sides of your skull) on both the left and right sides. Two of these spots at the very top of your head (the vertex) show a reaction in the outer layer of the bone called periosteal reaction, which means the outer bone layer is reacting to something underneath. These findings are not fully explained yet, so they are considered indeterminate.

Regarding your facial weakness, there is mild swelling and dye uptake in the part of the left facial nerve before it enters the ear (the preganglionic segment). This is consistent with Bell's palsy, which is a common condition that causes temporary weakness in the facial muscles. The radiologist’s overall impression is that there is no disease inside the brain itself. The bone spots are likely not dangerous tumors like glomus jugulare or lymphoma because they don't enhance with dye, but they do require further evaluation to rule out any malignant diseases that might involve the bone marrow.

To make sure everything is safe, two follow-up actions are recommended. First, you should have a CT scan of the temporal bones (the bones around the ears) as a routine next step. This will help doctors see the tiny internal structure of the bone, known as bone trabeculation, more clearly. Second, and more urgently, you need to be evaluated for any malignant diseases that could be causing these bone marrow changes. This is important to rule out any serious conditions given the presence of multiple bone spots with the outer bone reaction.

Review of Version A	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Simplified version B

This report is from a detailed brain and ear scan called an MRI (Magnetic Resonance Imaging), which uses strong magnets and radio waves to create clear pictures of the inside of your head without using any radiation. The scan was done to look at the cause of your left face and jaw pain and the weakness in your facial muscles that started a month ago.

First, the good news is that the main parts of your brain are healthy. The brainstem (the part that connects the brain to the spine and controls basic life functions), the cerebellum (the part of the brain that controls balance and coordination), and the main brain tissue all look normal. The fluid-filled spaces in the brain, called ventricles and sulci, are also normal. The scan did not find any signs of a stroke, bleeding, or tumors inside the brain itself. Additionally, the inner ear structures and the air spaces in the middle ear and mastoid bone (the area behind the ear) are clear and healthy.

Regarding your facial nerve, which controls the muscles in your face, the part of the nerve that runs through the ear bone looks normal and is not swollen. However, the part of the nerve just before it enters the ear shows mild swelling and inflammation. This finding is consistent with Bell's palsy, which is a common condition where the facial nerve becomes inflamed, often causing temporary weakness or paralysis on one side of the face. This explains your recent facial symptoms.

There are some findings in the bones of your skull that need attention. The doctors found a small, solid spot in the bone near the base of your skull on the left side, close to where the facial nerve runs. This spot does not take up the special dye used in the scan, which is a reassuring sign because it makes it less likely to be an aggressive tumor like a glomus jugulare (a type of vascular tumor) or lymphoma (a cancer of the immune system). This same type of spot was seen on your previous scans, and its size has not changed, suggesting it is stable. The doctors also found similar small spots in the top part of your skull on both sides. These spots are in the bone marrow (the spongy tissue inside the bone that makes blood cells). While the lack of dye uptake is a good sign, the doctors note that these bone changes should be evaluated to rule out any conditions that might affect the bone marrow.

Because of these bone findings, the radiologist recommends a CT scan (a type of X-ray that creates detailed cross-sectional images) of the temporal bone (the bone at the side of your head). This additional test will help the doctors see the internal structure of the bone more clearly to determine exactly what is happening with these spots. Please follow up with your doctor to schedule this recommended test and discuss the next steps for your care.

Review of Version B	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Original report

258 words. Followed by 2 simplified versions.

BRAIN (CRANIAL) MRI

Clinical information: Headache

Technique: Sagittal and axial T1-weighted; sagittal, axial and coronal FLAIR; axial T2-weighted, DWI, ADC and SWI. (20 ml Dotarem) 3-plane T1-weighted after contrast agent injection.

Findings:

Abscesses are observed in the left parietal lobe, posterior periventricularly located and adjacent to each other, the posterior one measuring approximately 20x18 mm, and the anterior one periventricularly located measuring 18x8 mm and opening into the ventricle, with walls showing contrast enhancement and diffusion restriction in the center. Contrast enhancement is observed in the ependyma extending from the occipital atrium to the body portion of the left lateral ventricle at the ventricle and opening site, and abscess foci showing diffusion restriction are observed within the ventricle and in the occipital horns of both ventricles. Thickening and signal increase not showing contrast enhancement are present in the ependyma in the walls of the right lateral ventricle and 3rd and 4th ventricles and in the cerebral aqueduct. Ventricular system size is normal. Vasogenic edema extending to the corpus callosum is observed in the white matter of the parietal and occipital lobes. Hemorrhage was not detected.

Aside from this, bilateral basal ganglia, thalamus, brainstem, and cerebellum appear normal.

Main cerebral vascular structures are patent.

Cranial bone structures, both globes, retrobulbar fat tissue, optic nerves, and optic chiasm are normal.

Impression:

· Cerebral abscesses in the left parietal lobe, posterior periventricular location, opening into the ventricle, ventriculitis
· Mucosal thickening in the bilateral maxillary sinuses, ethmoidal cells, and frontal sinuses.
· Minimal inflammatory signal increases in the right mastoid cells.

Simplified version A

You had a cranial MRI, which is a detailed brain scan using magnetic fields and radio waves without any radiation. To get the clearest picture, the doctors used several different types of images called sequences, including T1-weighted, FLAIR, T2-weighted, DWI, ADC, and SWI. They also injected a contrast dye called Dotarem to help highlight specific areas. The reason for this scan was to investigate your headache.

First, it is good news that many important parts of your head and brain look healthy. The size of the ventricular system (the fluid-filled spaces in the brain) is normal. The bilateral basal ganglia (deep brain structures involved in movement), thalamus (a relay station for sensory signals), brainstem (the part connecting the brain to the spine), and cerebellum (the part of the brain that controls balance and coordination) all appear normal. Your main cerebral vascular structures (the major blood vessels in the brain) are open and working properly. The cranial bone structures (your skull), both globes (your eyeballs), retrobulbar fat tissue (the fat behind the eyes), optic nerves (the nerves carrying vision signals), and optic chiasm (where the optic nerves cross) are all normal as well.

However, the scan did find some serious issues that need attention. There are two abscesses (pockets of infection and pus) in the left parietal lobe (the part of the brain involved in processing sensory information). The first one is in the posterior periventricular area (near the back of the fluid spaces) and measures approximately 20x18 mm. The second one is in the anterior periventricular area (near the front of the fluid spaces) and measures 18x8 mm. This second abscess is actually opening into the ventricle. The walls of both abscesses light up with the contrast dye, and the centers show what is called diffusion restriction, which confirms they are acute (new) and active infections.

Because these abscesses are connected to the fluid spaces, there is inflammation in the ependyma (the thin lining of the ventricles). This inflammation extends from the occipital atrium (a specific part of the ventricle) to the body of the left lateral ventricle. There are also signs of infection, known as abscess foci with diffusion restriction, inside the ventricle itself and in the occipital horns (the back parts) of both ventricles. This condition is called ventriculitis (inflammation of the ventricles). Additionally, the linings of the right lateral ventricle, the 3rd and 4th ventricles, and the cerebral aqueduct (the narrow channel connecting the 3rd and 4th ventricles) show thickening and increased signal, but they do not light up with the contrast dye. This finding is indeterminate, meaning it is not fully clear if it is active infection or something else, but it is not currently enhancing.

The infection has also caused swelling, called vasogenic edema, in the white matter (the nerve fibers) of the parietal and occipital (back of the brain) lobes. This swelling extends to the corpus callosum (the bridge of nerve fibers connecting the two halves of the brain). This swelling is acute (new).

Finally, the scan showed some minor issues in the sinuses and ears. There is mucosal thickening (swelling of the lining) in the bilateral maxillary sinuses (cheek sinuses), ethmoidal cells (sinuses between the eyes), and frontal sinuses (forehead sinuses). This is indeterminate and not currently causing enhancement. There are also minimal inflammatory signal increases in the right mastoid cells (the bone behind the ear), which suggests a mild, indeterminate inflammation there.

In summary, the radiologist's overall impression is that you have cerebral abscesses in the left parietal lobe that are opening into the ventricle, causing ventriculitis. There is also mucosal thickening in the maxillary, ethmoidal, and frontal sinuses, and minimal inflammation in the right mastoid cells. These findings are serious and require immediate medical attention from your doctor to treat the infection.

Review of Version A	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Simplified version B

This report is from a brain MRI, which is a detailed scan of the brain using magnetic fields and radio waves, without any radiation. The scan was performed to investigate your headache. The radiologist used special techniques and a contrast dye to get a very clear picture of the brain's structure and blood flow.

First, it is important to know that many parts of your head and brain are healthy. The bones of your skull, your eyes, the fat behind your eyes, and the optic nerves (the cables that carry vision from your eyes to your brain) are all normal. The main blood vessels in your brain are open and working correctly. Additionally, the basal ganglia (deep brain areas involved in movement), thalamus (a relay station for sensory signals), brainstem (the part connecting the brain to the spine), and cerebellum (the part of the brain that controls balance and coordination) all appear normal. There is no bleeding in the brain, and the overall size of the fluid-filled spaces in the brain is normal.

However, the scan did find some serious issues that need attention. There are brain abscesses, which are pockets of infection and pus, located in the left parietal lobe (the part of the brain that helps process touch and spatial awareness). These abscesses are near the fluid-filled spaces in the brain called ventricles. One of these abscesses is about the size of a small grape, and another is smaller. The walls of these abscesses light up with the contrast dye, which confirms they are active infections. More importantly, the infection has spread into the ventricles themselves. This condition is called ventriculitis, which means the lining of the fluid spaces in the brain is inflamed and infected. The scan shows signs of this inflammation in the left side and also some thickening of the lining in the right side and deeper parts of the brain.

Because of this infection, there is swelling in the brain tissue around the abscesses. This swelling, called vasogenic edema, has spread into the white matter (the communication pathways of the brain) of the parietal and occipital lobes (the back part of the brain involved in vision) and even into the corpus callosum (the bridge of nerve fibers connecting the left and right sides of the brain). This swelling is a reaction to the infection and is not a separate disease.

The report also notes some minor issues in the sinuses, which are the air-filled spaces in your face and forehead. There is some thickening of the lining in the maxillary sinuses (behind your cheeks), ethmoid cells (between your eyes), and frontal sinuses (above your eyebrows), which suggests a sinus infection or inflammation. There is also a tiny bit of inflammation in the mastoid cells, which are the spongy bones behind your ears.

In summary, the main finding is a brain infection with abscesses that have spread into the fluid spaces of the brain. This is a serious condition that requires immediate medical treatment, likely involving strong antibiotics and close monitoring by your doctors. The sinus and ear findings are less severe but may be related to the source of the infection. Please discuss these results with your healthcare provider right away to start the appropriate treatment.

Review of Version B	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Original report

133 words. Followed by 2 simplified versions.

BRAIN MRI (CRANIAL) [date_1]:
Clinical information: Headache, dizziness.
Technique: 3-plane T1-weighted, FLAIR; axial T2-weighted, SWI, and DWI.
Findings:
Apart from nonspecific T2 hyperintensity in the bilateral tegmental tracts in the pons, the brainstem and cerebellum, fourth ventricle, and posterior fossa cisterns are normal.
The bilateral basal ganglia, internal capsules, and corpus callosum are normal.
Apart from chronic gliotic foci <5 mm in diameter in the posterior periventricular and frontal white matter, the cerebral parenchyma is normal.
No specific findings on SWI sections.
The third and lateral ventricles and sulci are normal.
The major cerebral vascular structures are patent.
Left chronic otitis is present. The paranasal sinuses and right middle ear aeration, nasopharynx, and orbits are normal.
No specific findings in the visualized bones.
Impression:
• Chronic ischemic-gliotic lesions in the cerebral white matter.

Simplified version A

You had a brain MRI, which is a detailed scan of your head using magnetic fields and radio waves, with no radiation involved. This test was done to help investigate your headaches and dizziness. The good news is that most of your brain looks healthy. The brainstem (the part of the brain that connects to the spinal cord and controls basic functions like breathing), the cerebellum (the part of the brain that controls balance and coordination), and the fluid-filled spaces at the back of the skull all appear normal. The deep parts of the brain, including the basal ganglia (areas involved in movement control), the internal capsules (bundles of nerve fibers), and the corpus callosum (the bridge connecting the two halves of the brain), are also normal. The fluid spaces in the center of the brain, known as the ventricles, and the grooves on the brain's surface, called sulci, are the right size and shape. The major blood vessels in your brain are open and flowing properly, and the bones of your skull show no problems.

There are a few minor findings that are likely related to aging or past changes in blood flow. The report mentions "chronic ischemic-gliotic lesions" in the white matter (the inner wiring of the brain) near the front and back of the brain. These are tiny scar spots, smaller than 5 millimeters, caused by reduced blood flow in the past. They are very common, especially as people get older or if they have had high blood pressure, and they are not new or dangerous. There is also a slight brightness seen in the pons (a part of the brainstem), but this is described as nonspecific, meaning it is not a clear sign of a specific disease. Additionally, the scan shows signs of a long-standing ear infection in your left ear, called chronic otitis. However, your right ear, the sinuses (air pockets in the face bones), the throat area, and the eye sockets all look normal. The radiologist did not find any signs of bleeding, tumors, or acute strokes. No further tests are specifically recommended in this report, but you should discuss these minor findings with your doctor to see if any routine monitoring is needed.

Review of Version A	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Simplified version B

Your doctor ordered a brain MRI, which is a detailed scan of the brain using magnetic fields and radio waves, not radiation, to help investigate your headache and dizziness. The scan used several different types of images, including T1-weighted, FLAIR, T2-weighted, SWI, and DWI sequences, to look at the brain from many angles.

First, the good news is that most of your brain and surrounding areas look normal. The brainstem (the part of the brain that connects to the spinal cord and controls basic life functions), the cerebellum (the part of the brain that controls balance and coordination), the fourth ventricle (a fluid-filled space in the brain), and the posterior fossa cisterns (fluid-filled spaces at the back of the brain) are all normal. The bilateral basal ganglia (deep brain structures involved in movement), internal capsules (nerve pathways in the brain), and corpus callosum (the bridge connecting the two halves of the brain) are also normal. The third and lateral ventricles (fluid-filled spaces in the center of the brain) and the sulci (the grooves on the surface of the brain) are normal. The major cerebral vascular structures (the main blood vessels in the brain) are open and working properly. Additionally, the paranasal sinuses (air pockets around the nose), the right middle ear aeration (air in the right middle ear), the nasopharynx (the upper part of the throat behind the nose), the orbits (eye sockets), and the visualized bones all appear normal.

There are a few minor findings that your doctor will want to know about. In the bilateral tegmental tracts in the pons (specific nerve pathways in the lower part of the brainstem), there is a mild, nonspecific bright spot on the T2-weighted images. This is an indeterminate finding, meaning it is not specific to one disease and is often a common, minor variation that does not necessarily indicate a serious problem. In the posterior periventricular and frontal white matter (the inner tissue of the brain near the fluid spaces and at the front), there are chronic gliotic foci that are less than 5 mm in diameter. These are tiny, chronic scar spots in the brain caused by reduced blood flow in the past, often seen with aging or high blood pressure. These are described as chronic ischemic-gliotic lesions, which means they are old, small scars rather than new injuries. Finally, there is a moderate finding of chronic otitis in the left ear, which means there is a long-standing inflammation or infection in that ear.

The overall impression from the radiologist is that you have chronic ischemic-gliotic lesions in the cerebral white matter. No further specific tests or follow-ups were explicitly recommended in this report, but your doctor will review these findings with you to determine if any action is needed.

Review of Version B	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Original report

294 words. Followed by 2 simplified versions.

BRAIN MRI

Clinical information: Epilepsy

Technique: Sagittal and axial T1-weighted; sagittal, axial FLAIR; axial T2-weighted, DWI, ADC. Coronal thin sections perpendicular to the hippocampi T2-weighted, FLAIR, T1 IR.

Findings:

A cavernoma is observed in the left amygdala. Two cavernomas are observed in the left inferior frontal gyrus, parasagittal area of the left superior frontal gyrus, cingulate gyrus, and left precentral gyrus. Adjacent to the cavernoma in the left cingulate gyrus, in the parasagittal area, areas of chronic blood elements and sequela of prior hemorrhage are observed.

A resection cavity of the removed cavernoma is observed in the posterior part of the left temporal lobe, in the location of the inferior temporal gyrus. Chronic blood elements are present around the resection cavity.

The brainstem and basal ganglia appear normal. Vascular structures appear normal.

Signal intensity of other parenchymal areas is normal. Central and peripheral CSF spaces are of normal width. No space-occupying lesion was detected in the cavernous sinus, sella and suprasellar cisterns, or brainstem. An arachnoid cyst is observed in the posterior part of the cerebellum in the posterior fossa.

Retention cysts are observed in both maxillary sinuses.

No specific feature was detected in diffusion-ADC and SWI sections.

The third ventricle and both lateral ventricles are of normal width. Hemispheric cortical sulcal widths are compatible with the patient's age.

Major cerebral vascular structures are patent.

Bilateral orbits, paranasal sinuses, nasopharynx, and middle ear aeration are normal.

Visualized bony structures are normal.

Impression:

• Findings related to resection of the removed cavernoma from the left inferior temporal gyrus.
• Cavernoma foci in the left amygdala, frontal parasagittal gyrus, cingulate gyrus, inferior frontal gyrus, and precentral gyrus, sequela of prior parenchymal hemorrhage adjacent to the left cingulate (underlying cavernoma presence cannot be excluded at this level).

Simplified version A

This report is from a brain MRI, which is a detailed scan of the brain using magnetic fields and radio waves, without any radiation. The scan was performed to look for the cause of epilepsy (seizures). First, it is good to know that many parts of the brain are healthy. The brainstem (the part that connects the brain to the spine and controls basic life functions) and the basal ganglia (deep brain structures that help control movement) look normal. The major blood vessels in the brain are open and working properly. The fluid spaces around the brain, including the ventricles (fluid-filled chambers inside the brain), are of normal size. The bones of the skull, the eyes, the sinuses, and the middle ears all appear normal as well.

The main findings relate to small, abnormal blood vessel clusters called cavernomas. A cavernoma is a tangle of tiny, weak blood vessels that can sometimes leak small amounts of blood. The scan shows several of these cavernomas in the left side of the brain. Specifically, there is one in the left amygdala (a small area deep in the brain involved in emotions and memory). There are also cavernomas in the left inferior frontal gyrus (a part of the brain involved in speech and planning), the left superior frontal gyrus (involved in thinking and decision-making), the cingulate gyrus (involved in attention and emotion), and the left precentral gyrus (the

main area for controlling voluntary movement). Near the cavernoma in the cingulate gyrus, there are signs of old blood and scarring from a previous small bleed. This is a chronic finding, meaning it happened in the past and is not a new emergency.

The scan also shows the result of a previous surgery. There is a cavity in the back part of the left temporal lobe (the area of the brain involved in hearing and language) where a cavernoma was previously removed. This is called a resection cavity. There is some old blood and scarring around this surgical site, which is expected after such a procedure.

Additionally, the scan found a small, fluid-filled sac called an arachnoid cyst in the back of the cerebellum (the part of the brain that controls balance and coordination). This is a common, usually harmless finding. There are also small fluid-filled sacs, called retention cysts, in both maxillary sinuses (the air-filled spaces behind the cheeks). These are very common and often do not cause symptoms.

In summary, the scan confirms the presence of multiple cavernomas in the left side of the brain and shows the healing site from a previous surgery to remove one of them. The other parts of the brain and surrounding structures look normal. The findings are consistent with the patient's history of epilepsy and previous surgery. No new or urgent problems were found.

Review of Version A	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	

Simplified version B

This report is from a Brain MRI, which is a detailed scan of the brain using magnetic fields and radio waves, without any radiation. The scan was performed to investigate epilepsy (a condition causing seizures). The radiologist used several specific types of images, or sequences, to look at the brain in detail, including T1-weighted, FLAIR, T2-weighted, DWI, ADC, T1 IR, and SWI.

First, it is good news that many important parts of the brain and head are normal. The brainstem (the part of the brain that connects to the spinal cord and controls basic life functions) and the basal ganglia (deep structures in the brain that help control movement) appear normal. The vascular structures (blood vessels) are normal, and the other areas of brain tissue (parenchyma) look healthy. The central and peripheral spaces filled with fluid (CSF) are of normal width. There are no tumors or masses in the cavernous sinus (a space at the base of the brain), the sella (the bony pocket holding the pituitary gland), the suprasellar cisterns (fluid spaces above the pituitary), or the brainstem. The third ventricle and both lateral ventricles (fluid-filled spaces in the center of the brain) are of normal width. The spaces between the folds of the brain (cortical sulci) are normal for the patient's age. The major cerebral vascular structures are open and working properly. The

bilateral orbits (eye sockets), paranasal sinuses (air pockets in the face bones), nasopharynx (the upper part of the throat behind the nose), and middle ear aeration (air in the middle ear) are all normal. The visualized bony structures are also normal.

However, the scan did find some abnormal areas. A cavernoma is a small, tangle of weak blood vessels that can leak blood. This is a chronic (long-standing) condition. The scan found a cavernoma in the left amygdala (a part of the brain involved in emotion and memory). There are also two cavernomas in the left inferior frontal gyrus (a part of the brain involved in speech and planning). Additionally, there is a cavernoma in the parasagittal area of the left superior frontal gyrus (the top front part of the brain), a cavernoma in the left cingulate gyrus (a part of the brain involved in attention and emotion), and a cavernoma in the left precentral gyrus (the area just in front of the motor strip, which controls movement). Near the cavernoma in the left cingulate gyrus, there are areas of old blood (chronic blood elements) and signs of a past bleed (sequela of prior hemorrhage). It is not possible to rule out another cavernoma right next to this area based on this scan.

There is also a resection cavity (a hole left after surgery) in the posterior part of the left temporal lobe (the side of the brain involved in hearing and language), specifically in the location of the inferior temporal gyrus. This is where a cavernoma was previously removed. There are also old blood elements around this surgical cavity.

There are two incidental findings, which are things found by chance that are usually not related to the main problem. There is an arachnoid cyst (a harmless fluid-filled sac) in the posterior part of the cerebellum (the part of the brain that controls balance and coordination) in the posterior fossa (the lower back part of the skull). There are also retention cysts (small, harmless fluid-filled sacs) in both maxillary sinuses (the sinuses behind the cheeks).

In summary, the main findings are related to the previous surgery to remove a cavernoma from the left inferior temporal gyrus. There are also several other cavernomas in the left amygdala, frontal parasagittal gyrus, cingulate gyrus, inferior frontal gyrus, and precentral gyrus, along with signs of a past bleed near the left cingulate area.

Review of Version B	
Accuracy	☐ No errors ☐ Minor errors ☐ Serious errors
Invented information	☐ None ☐ Yes (please note below)
Important information missing	☐ None ☐ Yes (please note below)
Easy for a patient or family to understand	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = very hard, 5 = very easy)
Could this be given to the patient?	☐ As written ☐ After edits ☐ No
What was done well / what was missed or wrong	